第四章：电磁波的传播

前言

- 在迅变情况下，电磁场会相互激发，电磁场以波动形式存在，形成空间传播的电磁波

- 电磁波的传播、辐射和激发问题已发展为独立的学科，具有十分丰富的内容

本章先不关注电磁波是如何产生的，只专注于介绍关于电磁波传播的基本理论知识

目录

- 平面电磁波
- 介质、导体的频率色散
- 导体存在时的电磁波
- 电磁波在各向异性介质中的传播
- 旋光介质中的电磁波
- 电磁波在界面的反射与折射
- 波导
- 谐振腔

平面电磁波

1. 电磁场的波动方程

$$\begin{cases} \nabla \cdot \vec{D} = \rho_f \\ \nabla \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t} \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{H} = \vec{J}_f + \dfrac{\partial \vec{D}}{\partial t} \end{cases}$$

$\rho_f = 0,\ \vec{J}_f = 0$



无源情况下电磁场的运动

$$\begin{cases} \nabla \cdot \vec{D} = 0 \\ \nabla \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t} \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{H} = \dfrac{\partial \vec{D}}{\partial t} \end{cases}$$

- 先考虑真空中的电磁波：$\vec{D} = \varepsilon_0 \vec{E}$，$\vec{B} = \mu_0 \vec{H}$，取：

$$\nabla \times (\nabla \times \vec{E}) = \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\varepsilon_0 \mu_0 \frac{\partial}{\partial t}\left(\frac{\partial \vec{E}}{\partial t}\right)$$

平面电磁波

$$\nabla^2\vec{E}-\varepsilon_0\mu_0\frac{\partial^2}{\partial t^2}\vec{E}=0$$

$$\varepsilon_0\mu_0=\frac{1}{c^2}$$

同理可得：

$$\nabla^2\vec{B}-\varepsilon_0\mu_0\frac{\partial^2}{\partial t^2}\vec{B}=0$$

三维波动方程

$$(\nabla^2-\frac{\partial^2}{c^2\partial t^2})\begin{pmatrix}\vec{E}\\ \vec{B}\end{pmatrix}=0$$

光速，最重要的物理常数之一

$$c=\frac{1}{\sqrt{\varepsilon_0\mu_0}}$$

- 介质情况比较复杂，如果考虑线性无色散介质中$\vec{D}=\varepsilon\vec{E}$，$\vec{B}=\mu\vec{H}$

三维波动方程

$$(\nabla^2-\varepsilon\mu\frac{\partial^2}{\partial t^2})\begin{pmatrix}\vec{E}\\ \vec{B}\end{pmatrix}=0$$

波速 $v=\frac{1}{\sqrt{\varepsilon\mu}}$

2. 时谐电磁波

电磁波往往以确定的频率振荡，若不是，可以做傅里叶分析，分解为不同频率的正弦波的叠加

- 场量的复数表示

对于一特定频率的电磁波，电磁场频率为ω，电磁场对时间的依赖关系为$\cos\omega t$，可以采用复数表示

$$\vec{E}_0\cos(\omega t+\varphi)=\mathrm{Re}\{\vec{E}_0e^{-i(\omega t+\varphi)}\}=\mathrm{Re}\{\vec{E}_0e^{-i\varphi}e^{-i\omega t}\}$$

$$\vec{E}_0e^{-i\varphi}\rightarrow\vec{E}$$

真实物理场，实数　　复数场$\vec{E}\,e^{-i\omega t}$

$$\vec{E}(\vec{x},t)=\vec{E}(\vec{x})\,e^{-i\omega t}$$

$$\vec{B}(\vec{x},t)=\vec{B}(\vec{x})\,e^{-i\omega t}$$

平面电磁波

➢ 时谐情形下的Maxwell方程组

$$\begin{cases} \nabla\cdot\vec{D}=0 \\ \nabla\times\vec{E}=-\dfrac{\partial\vec{B}}{\partial t} \\ \nabla\cdot\vec{B}=0 \\ \nabla\times\vec{H}=\dfrac{\partial\vec{D}}{\partial t} \end{cases}$$

$$\vec{D}=\varepsilon\vec{E},\ \ \vec{B}=\mu\vec{H}$$

$$\vec{E}(\vec{x},t)=\vec{E}(\vec{x})\,e^{-i\omega t}$$

$$\vec{B}(\vec{x},t)=\vec{B}(\vec{x})\,e^{-i\omega t}$$

$$\begin{cases} \nabla\cdot\vec{E}=0 \\ \nabla\times\vec{E}=i\omega\vec{B} \\ \nabla\cdot\vec{B}=0 \\ \nabla\times\vec{H}=-i\omega\varepsilon\vec{E} \end{cases}$$

$$\frac{\partial}{\partial t}\to -i\omega$$

取第二、四式旋度

$$(\nabla^2-\varepsilon\mu\frac{\partial^2}{\partial t^2})\begin{pmatrix}\vec{E}(\vec{x},t)\\ \vec{B}(\vec{x},t)\end{pmatrix}=0$$

$$\frac{\partial^2}{\partial t^2}\to -\omega^2$$

$$(\nabla^2+\omega^2\varepsilon\mu)\begin{pmatrix}\vec{E}(\vec{x})\\ \vec{B}(\vec{x})\end{pmatrix}=0$$

取 $k^2=\omega^2\varepsilon\mu$

$$(\nabla^2+k^2)\begin{pmatrix}\vec{E}(\vec{x})\\ \vec{B}(\vec{x})\end{pmatrix}=0$$ 亥姆霍兹方程

平面电磁波

由此可得，一定频率ω下的Maxwell方程组：

$$\begin{cases}(\nabla^2+k^2)\vec{E}(\vec{x})=0\\ \nabla\cdot\vec{E}=0\\ \vec{B}=\dfrac{1}{i\omega}\nabla\times\vec{E}\end{cases}\quad 或\quad \begin{cases}(\nabla^2+k^2)\vec{B}(\vec{x})=0\\ \nabla\cdot\vec{B}=0\\ \vec{E}=\dfrac{i}{\omega\mu\varepsilon}\nabla\times\vec{B}\end{cases}$$

- 色散关系,电磁波频率与波矢之间的关系，由 $k^2=\omega^2\varepsilon\mu$，可得

$$k=\pm\,\omega\sqrt{\varepsilon\mu}\qquad \text{or}\qquad k=\pm\,\omega\sqrt{\varepsilon}\sqrt{\mu}$$

- 亥姆霍兹方程的有多种形式，如平面波、球面波、高斯光束、贝塞尔光束、艾利光束等，当然最简单的自然是平面波。

平面电磁波

3. 平面电磁波

平面电磁波，即在垂直于波传播方向的正交面上各点具有相同的值，即波阵面或者等相位面为平面的电磁波。

$$(\nabla^2 + k^2)\vec{E}(\vec{x}) = 0$$

$$\vec{E}(\vec{x}) = \vec{E}_0 e^{i\vec{k}\cdot\vec{x}} \quad \Leftarrow \quad \left(\frac{d^2}{dx^2} + k^2\right)\vec{E}(\vec{x}) = 0$$

色散关系 $k = \omega\sqrt{\varepsilon\mu}$

同时利用$\nabla \cdot \vec{E} = 0$，可知$i\vec{k} \cdot \vec{E}_0 = 0$, 电场与波传播的方向垂直，平面电磁波

$$\vec{E}(\vec{x}, t) = \vec{E}_0 e^{i(\vec{k}\cdot\vec{x} - \omega t)}$$

$\vec{E}_0$振幅　$e^{i(\vec{k}\cdot\vec{x} - \omega t)}$相位因子　　$\vec{k}$波矢

实际物理场，应该是实数, 即$\vec{E}(\vec{x}, t) = \mathrm{Re}\{\vec{E}_0 e^{i(\vec{k}\cdot\vec{x} - \omega t)}\}$

平面电磁波

- 相速度，即等相位面移动的速度。由相位$\vec{k}\cdot\vec{x}-\omega t$，可知

$$k\Delta x-\omega\Delta t=0 \qquad v_p=\frac{\Delta x}{\Delta t}=\frac{\omega}{k}$$

- 波长，$k\lambda=2\pi$, 可得波长$\lambda=\frac{2\pi}{k}$

- 周期，$\omega T=2\pi$, 可得周期$T=\frac{2\pi}{\omega}$，频率$f=\frac{1}{T}=\frac{\omega}{2\pi}$, 角频率$\omega=2\pi f$

- 平面电磁波电磁场方向、大小关系

$$\begin{cases}\vec{E}(\vec{x},t)=\vec{E}_0e^{i(\vec{k}\cdot\vec{x}-\omega t)}\\ \vec{k}\cdot\vec{E}_0=0\\ \vec{B}=\frac{1}{i\omega}\nabla\times\vec{E}\Rightarrow\vec{B}_0=\frac{1}{\omega}\vec{k}\times\vec{E}_0\end{cases}$$

$\vec{k}$垂直于$\vec{E}$, $\vec{B}$,并且$(\vec{k},\vec{E},\vec{B})$构成右手定则

平面电磁波为横波

平面电磁波

电磁场振幅比值：

$$\frac{|\vec{E}_0|}{|\vec{B}_0|}=\frac{|\vec{E}_0|}{|\frac{1}{\omega}\vec{k}\times\vec{E}_0|}=\frac{\omega}{|\vec{k}|}=v_p$$

真空中，SI单位制下比值为c

$$v_p=\frac{\omega}{k}=\frac{1}{\sqrt{\mu\varepsilon}}$$

$$\frac{|\vec{E}_0|}{|\vec{H}_0|}=\frac{|\vec{E}_0|}{|\frac{1}{\omega\mu}\vec{k}\times\vec{E}_0|}=\frac{\mu\omega}{|\vec{k}|}=\mu v_p=\sqrt{\frac{\mu}{\varepsilon}}=Z$$

$Z=\sqrt{\frac{\mu}{\varepsilon}}$ 阻抗

$$Z_0=\sqrt{\frac{\mu_0}{\varepsilon_0}}=376.7\Omega$$

平面电磁波的特性总结：

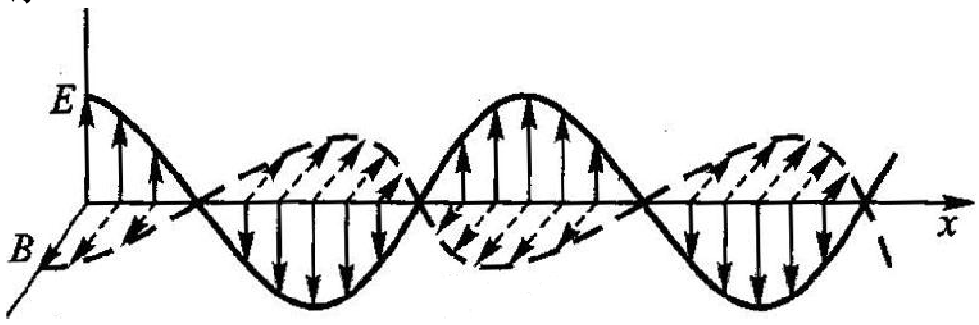
（1）平面电磁波为横波，$\vec{E}$和$\vec{B}$都与传播方向垂直

（2）$\vec{E}$和$\vec{B}$互相垂直，$(\vec{k},\vec{E},\vec{B})$构成右手定则

（3）$\vec{E}$和$\vec{B}$的振幅比为v_p，同相位

$k=\omega\sqrt{\varepsilon\mu}$ 乘积

$Z=\sqrt{\frac{\mu}{\varepsilon}}$ 比值



平面电磁波

4. 平面电磁波的偏振

设平面电磁波沿z方向传播，则有$\hat{z}\cdot\vec{E}_0=0,\hat{z}\cdot\vec{B}_0=0$，因此电、磁场在x-y平面内

可设为：$\vec{E}_0=E_{0x}\hat{x}+E_{0y}\hat{y}$

$=|E_{0x}|e^{i\phi_x}\hat{x}+|E_{0y}|e^{i\phi_y}\hat{y}$

- 若$\phi_x=\phi_y$，线偏振
- 若$\phi_x\neq\phi_y$，椭圆偏振

特例：$|E_{0x}|=|E_{0y}|=E_0$，但是$\phi_y=\phi_x\pm\frac{\pi}{2}$

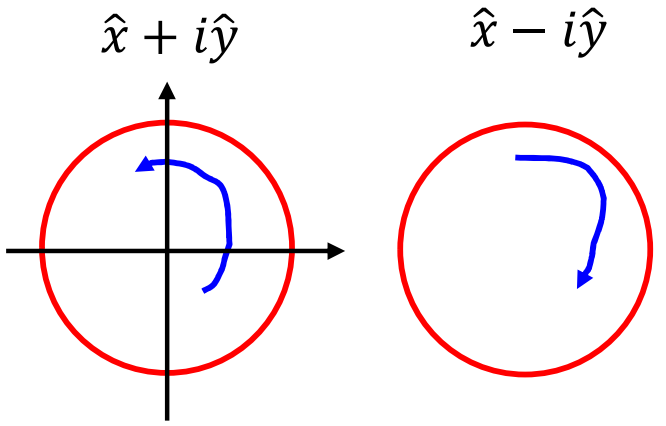
$\vec{E}=(E_0\hat{x}+E_0\hat{y})e^{i(kz-\omega t)}$

$=E_0(\hat{x}\pm i\hat{y})\ e^{i(kz-\omega t)}$

真实的物理场：$E_x=E_0\cos(kz-\omega t)$

$E_y=\mp E_0\sin(kz-\omega t)$

在空间中一点，迎着电磁波看电矢量在时间上的轨迹



左旋偏振光　　右旋偏振光

对于已经给定的电磁波，如何得到其偏振态？

平面电磁波

5. 电磁波的能量、能流

电磁场的能量密度 $w = \frac{1}{2}\left(\vec{E}\cdot\vec{D} + \vec{H}\cdot\vec{B}\right)$

$$= \frac{1}{2}\left(\varepsilon E^2 + \frac{1}{\mu}B^2\right)$$

$\vec{D} = \varepsilon\vec{E}$，$\vec{B} = \mu\vec{H}$

- 对于平面电磁波 $E^2 = v_p^2 B^2 = \frac{1}{\varepsilon\mu}B^2$, 因此$\varepsilon E^2 = \frac{1}{\mu}B^2$，电场、磁场能量相等；
- 能流密度$\vec{S} = \vec{E}\times\vec{H} = \vec{E}\times\frac{1}{Z}\left(\hat{k}\times\vec{E}\right) = \frac{1}{Z}E^2\hat{k} = \frac{1}{\sqrt{\varepsilon\mu}}(\varepsilon E^2)\hat{k} = w\vec{v}$
- 时间平均的能量、能流

流密度=体密度*流速

能量、能流等是**电磁场的多次式**，其计算不能直接用复数

平面电磁波

瞬时能量密度

$$w = \varepsilon E_0^2 \cos^2\left(\vec{k}\cdot\vec{x} - \omega t\right) = \frac{1}{2}\varepsilon E_0^2[1 + \cos 2(\vec{k}\cdot\vec{x} - \omega t)]$$

平均能量密度

$$\langle w\rangle = \frac{1}{T}\int_0^T \frac{1}{2}\varepsilon E_0^2[1 + \cos 2(\vec{k}\cdot\vec{x} - \omega t)]\, dt$$

$$= \frac{1}{2}\varepsilon E_0^2$$

$$T = \frac{2\pi}{\omega}$$

*二次式求平均值的一般公式 $f(t) = f_0 e^{-i\omega t}$ $g(t) = g_0 e^{-i\omega t} e^{i\phi}$

$$\langle fg\rangle = \frac{\omega}{2\pi}\int_0^{\frac{2\pi}{\omega}} dt f_0 \cos\omega t\, g_0 \cos(\omega t - \phi)$$

$$= \frac{\omega}{2\pi}\int_0^{\frac{2\pi}{\omega}} dt f_0 g_0 \frac{1}{2}\cos\phi = \frac{1}{2} f_0 g_0 \cos\phi \quad = \frac{1}{2} Re\{f_0 g_0^*\}$$

$$\langle fg\rangle = \frac{1}{2} Re\{f_0 g_0^*\}$$

平面电磁波

平均能流密度

$$\vec{H} = \frac{1}{Z}(\hat{k} \times \vec{E})$$

$$\langle \vec{S} \rangle = \langle \vec{E} \times \vec{H} \rangle = \frac{1}{2}\mathrm{Re}\{\vec{E} \times \vec{H}^*\}$$

$$= \frac{1}{Z}E_0^2\hat{k} = \frac{1}{\sqrt{\mu\varepsilon}}(\varepsilon E_0^2)\hat{k}$$

$$= \langle w \rangle \vec{v}$$

流密度=体密度*流速

6. 电磁波的动量、动量流密度

- 电磁波的动量

$$\vec{g} = \varepsilon_0 \vec{E} \times \vec{B} = \varepsilon_0 \mu_0 \frac{1}{Z_0}\vec{E} \times (\hat{k} \times \vec{E})$$

$$= \varepsilon_0 \mu_0 \frac{1}{Z_0}\vec{E} \times (\hat{k} \times \vec{E}) = \frac{1}{c}(\varepsilon_0 E^2)\hat{k} = \frac{w}{c}\hat{k}$$

能量/动量=光速

平面电磁波

- 电磁波的流密度

$$\overleftrightarrow{T}=\frac{1}{2}\left(\varepsilon_0 E^2+\frac{1}{\mu_0}B^2\right)\overleftrightarrow{I}-\varepsilon_0\vec{E}\vec{E}-\frac{1}{\mu_0}\vec{B}\vec{B}$$

$$\vec{E}\cdot\overleftrightarrow{T}=\frac{1}{2}\left(\varepsilon_0 E^2+\frac{1}{\mu_0}B^2\right)\vec{E}-\varepsilon_0 E^2\vec{E}=0$$

$$\overleftrightarrow{T}\cdot\vec{E}=\vec{B}\cdot\overleftrightarrow{T}=\overleftrightarrow{T}\cdot\vec{B}=0$$

$$\hat{k}\cdot\overleftrightarrow{T}=\frac{1}{2}\left(\varepsilon_0 E^2+\frac{1}{\mu_0}B^2\right)\hat{k}=\varepsilon_0 E^2\hat{k}=w\hat{k}$$

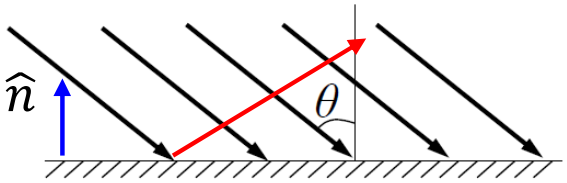
$$\overleftrightarrow{T}\cdot\hat{k}=w\hat{k}$$

$$\overleftrightarrow{T}=w\hat{k}\hat{k}=cg\hat{k}\hat{k}$$

流密度=体密度*流速

平面电磁波

例1：辐射压力，设平面电磁波入射于理想导体表面上而被全部反射，入射角为θ，求导体表面所受到的辐射压强。



电磁场切向分量无贡献，考虑法向分量：

$$\vec{F}\cdot\hat{n}=(-\oint\overleftrightarrow{T}\cdot d\vec{S})\cdot\hat{n}$$

$$\overleftrightarrow{T}=cg\hat{k}\hat{k}$$

$$\hat{k}\cdot\hat{n}=-\cos\theta$$

$$=-\oint cg\cos^2\theta dS$$

导体面所受的压强 $$P=\frac{F_n}{\oint dS}=2gc\cos^2\theta$$

压强的时间平均 $$\langle P\rangle=2\langle gc\rangle\cos^2\theta=2\langle w_i\rangle\cos^2\theta$$

w_i入射电磁波的能量

导体外部的电场 $$\vec{E}=\vec{E}_i+\vec{E}_r \qquad E^2=E_i^2+E_r^2+2\mathrm{Re}\{\vec{E}_i^*\cdot\vec{E}_r\}$$

$$\langle w\rangle=\langle w_i\rangle+\langle w_r\rangle=2\langle w_i\rangle$$

干涉项无贡献

平面电磁波

导体表面总的压强　　$\langle P\rangle = \langle w\rangle \cos^2\theta$

若电磁波从各个角度入射，需要对角度进行平均：

$$\langle P\rangle = \frac{1}{2\pi}\int \langle w\rangle \cos^2\theta d\Omega = \frac{1}{2\pi}\int_0^{2\pi} d\phi \int_0^{\pi/2} \langle w\rangle \cos^2\theta \sin\theta \, d\theta$$

$$= \frac{1}{2\pi}\int_0^{2\pi} d\phi \int_0^{\pi/2} \langle w\rangle \cos^2\theta \sin\theta \, d\theta = \frac{1}{3}\langle w\rangle$$

- 若表面完全吸收电磁波，黑体辐射对界面的压强，仍然满足
- 太阳照射在地球表面的能流密度约为1.35*10^3W/m^2，能量密度约为1.35*10^3W/m^2/c；压强约为~10^{-6}P。

介质、导体的频率色散

1. 色散介质的电磁响应

介质处于电磁场中，产生极化、磁化，其响应可以一般的描述为：

$$\vec{D}(\vec{r},t)=\iint \varepsilon(t-t',\vec{r}-\vec{r}')\vec{E}(\vec{r}',t')dt'd\vec{r}'$$

$$\vec{B}(\vec{r},t)=\iint \mu(t-t',\vec{r}-\vec{r}')\vec{H}(\vec{r}',t')dt'd\vec{r}'$$

- 时间、空间平移不变性质
- 因果律

考虑局域近似$\delta(\vec{r}-\vec{r}')$

$$\vec{D}(\vec{r},t)=\int_{-\infty}^{t} \varepsilon(t-t',\vec{r})\vec{E}(\vec{r},t')dt'$$

$$\vec{B}(\vec{r},t)=\int_{-\infty}^{t} \mu(t-t',\vec{r})\vec{H}(\vec{r},t')dt'$$

介质、导体的频率色散

考虑时谐场 $\vec{E}(\vec{r},\ t)=\vec{E}(\vec{r})e^{-i\omega t}$, $\vec{D}(\vec{r},\ t)=\vec{D}(\vec{r})e^{-i\omega t}$

$$\vec{D}(\vec{r},t)=\int_{-\infty}^{t}\varepsilon(t-t',\vec{r})\vec{E}(\vec{r},t')dt'$$

$$=\int_{-\infty}^{t}\varepsilon(t-t',\vec{r})\vec{E}(\vec{r})e^{-i\omega t'}dt'$$

$$=\int_{-\infty}^{t}\varepsilon(t-t',\vec{r})e^{i\omega t}e^{-i\omega t'}dt'\,\vec{E}(\vec{r})e^{-i\omega t}$$

$$=\int_{-\infty}^{t}\varepsilon(t-t',\vec{r})e^{-i\omega(t'-t)}d(t'-t)\,\vec{E}(\vec{r})e^{-i\omega t}$$

$$=\varepsilon(\omega)\vec{E}(\vec{r})e^{-i\omega t}=\vec{D}(\vec{r})e^{-i\omega t}$$

受迫振动

$$\Longrightarrow\quad \vec{D}(\vec{r})=\varepsilon(\omega)\vec{E}(\vec{r})$$

同理 $\vec{B}(\vec{r})=\mu(\omega)\vec{H}(\vec{r})$

- 特定频率下：色散介质的波动方程与真空、无色散介质形式一致

$$\varepsilon\rightarrow\varepsilon(\omega)$$
$$\mu\rightarrow\mu(\omega)$$

介质、导体的频率色散

2. 介电系数$\varepsilon(\omega)$的模型

考虑一电子（电荷-e）在原子荷束缚，做简谐振动，电磁场作用下：

$$m\left(\ddot{\vec{x}}+\gamma\dot{\vec{x}}+\omega_0^2\vec{x}\right)=-e\vec{E}(\vec{x},\,t)$$

磁力远小于静电力，因此忽略

所有物理量以$e^{-i\omega t}$谐振

$$m(-\omega^2-i\gamma\omega+\omega_0^2)\vec{x}=-e\vec{E}(\vec{x})$$

可得：$\vec{x}=\dfrac{-e\vec{E}(\vec{x})}{m(-\omega^2-i\gamma\omega+\omega_0^2)}$

该电子的偶极矩：

$$\vec{p}=-e\vec{x}=\frac{e^2\vec{E}(\vec{x})}{m(-\omega^2-i\gamma\omega+\omega_0^2)}$$

介质、导体的频率色散

假设物质由分子组成，其分子数密度N，每个分子贡献Z个电子，其中f_j个电子束缚频率ω_j，衰减常数γ_j

电位移矢量:

$$\vec{D} = \varepsilon\vec{E} = \varepsilon_0\vec{E} + \vec{P}$$

$$= \varepsilon_0\vec{E} + \frac{Ne^2}{m}\sum_j \frac{f_j\vec{E}}{-\omega^2 - i\gamma_j\omega + \omega_j^2}$$

$$\vec{P} = \frac{Ne^2}{m}\sum_j \frac{f_j\vec{E}}{-\omega^2 - i\gamma_j\omega + \omega_j^2}$$

$$\sum_j f_j = Z$$

振子强度

由此可得相对介电系数

$$\frac{\varepsilon}{\varepsilon_0} = 1 + \frac{Ne^2}{m\varepsilon_0}\sum_j \frac{f_j}{-\omega^2 - i\gamma_j\omega + \omega_j^2}$$

相对介电系数； 复数

介质、导体的频率色散

$$\frac{\varepsilon}{\varepsilon_0} = 1 + \frac{Ne^2}{m\varepsilon_0}\sum_j \frac{f_j}{-\omega^2 - i\gamma_j\omega + \omega_j^2}$$ 相对介电系数

- 一般情况下$\gamma_j \le \omega_j$， 高频下$\omega_j^2 - \omega^2$的贡献为负$\varepsilon/\varepsilon_0 < 1$，低频率 $\varepsilon/\varepsilon_0 > 1$；

当$\omega \approx \omega_j$, 实部为0，主要贡献纯虚数；

- 正常色散，$\varepsilon(\omega)$的实部随着ω增加而增大；反常色散，$\varepsilon(\omega)$的虚部占主要部分；
- 复介电系数下的电磁波，波数不再是实数，而是复数；

$$k = \beta + i\frac{\alpha}{2}$$

$$k^2 = \omega^2\varepsilon\mu = k_0^2\frac{\varepsilon}{\varepsilon_0} \Rightarrow \begin{cases} \beta^2 - \dfrac{\alpha^2}{4} = k_0^2\mathrm{Re}\{\dfrac{\varepsilon}{\varepsilon_0}\} \\ \alpha\beta = k_0^2\mathrm{Im}\{\dfrac{\varepsilon}{\varepsilon_0}\} \end{cases}$$

如果$\beta \gg \alpha$ 吸收系数

$$\frac{\beta}{\alpha} \approx \frac{\mathrm{Re}\{\varepsilon\}}{\mathrm{Im}\{\varepsilon\}}$$

平面电磁波 $\vec{E}(\vec{x},t) = \vec{E}_0 e^{ikx}e^{-i\omega t} = \vec{E}_0 e^{-\alpha x/2}e^{i\beta x}e^{-i\omega t}$

衰减

介质、导体的频率色散

$$\frac{\varepsilon}{\varepsilon_0} = 1 + \frac{Ne^2}{m\varepsilon_0}\sum_j \frac{f_j}{-\omega^2 - i\gamma_j\omega + \omega_j^2}$$

- 若在低频下，部分的电子是自由的，将这部分的电子的贡献单独考虑

$$\frac{\varepsilon}{\varepsilon_0} = 1 + \frac{Ne^2}{m\varepsilon_0}\sum_{j\neq 0} \frac{f_j}{-\omega^2 - i\gamma_j\omega + \omega_j^2} + \frac{Ne^2}{m\varepsilon_0}\frac{f_0}{-\omega^2 - i\gamma_0\omega} \qquad \omega_j\big|_{j=0} = 0$$

$$= \frac{\varepsilon_b}{\varepsilon_0} + \frac{iNe^2 f_0}{m\varepsilon_0}\frac{1}{\omega(\gamma_0 - i\omega)}$$

$Nf_0 \to n$ 自由电子数密度

$$\varepsilon = \varepsilon_b + \frac{ine^2}{m}\frac{1}{\omega(\gamma_0 - i\omega)}$$

$\frac{\varepsilon_b}{\varepsilon_0} \to 1$

$\omega_p^2 = \frac{ne^2}{m\varepsilon_0}$ 等离子频率

$$\varepsilon_r = \frac{\varepsilon_b}{\varepsilon_0} + \frac{ine^2}{m\varepsilon_0}\frac{1}{\omega(\gamma_0 - i\omega)} = 1 - \frac{\omega_p^2}{\omega(\omega + i\gamma_0)}$$

Drude 模型（1900）

介质、导体的频率色散

$$\varepsilon = \varepsilon_b + \frac{ine^2}{m}\frac{1}{\omega(\gamma_0 - i\omega)}$$

$$\varepsilon_r = 1 + \frac{i\sigma}{\omega\varepsilon_0}$$

$$\nabla \times \vec{H} = \vec{J}_f + \frac{\partial \vec{D}}{\partial t} = \sigma\vec{E} - i\omega\varepsilon_b\vec{E} = -i\omega\left(\varepsilon_b + \frac{i\sigma}{\omega}\right)\vec{E} = -i\omega\varepsilon\vec{E}$$

可以得到电导率 $\sigma = \frac{ne^2}{m}\frac{1}{\gamma_0 - i\omega}$

- 低频$\omega \to 0$, $\sigma_c = \frac{ne^2}{m\gamma_0}$，直流电导，与电子密度成正比，与衰减率程反比
- 电导率为复数，由欧姆损耗公式$\frac{1}{2}\mathrm{Re}\{\vec{E}\cdot\vec{J}^*\}$, 可知道电导率实部代表损耗，虚部代表能量转移
- 对于导体$\gamma_0 \approx 10^{13}\sim10^{14}$Hz, 可见在微波波段(GHz)波段实部占主导（$\sigma \to \sigma_c$）；在可见光波段($10^{15}$Hz)，电导率虚部占主要部分$\sigma \to i\frac{ne^2}{m\omega}$;
- 高频波段的金属，与介质区别不大，都是用复介电系数描述。

介质、导体的频率色散

金属的相对介电系数

$$\varepsilon_r = 1 - \frac{\omega_p^2}{\omega(\omega + i\gamma_0)}$$ Drude 模型（1900） $$\omega_p^2 = \frac{ne^2}{m\varepsilon_0}$$

- 低频，当$\omega \ll \omega_p$，$\omega \ll \gamma_0$

$$\varepsilon_r = 1 - \frac{\omega_p^2}{\omega(\omega + i\gamma_0)} = 1 - \frac{\omega_p^2}{\omega^2\left(1 + \frac{i\gamma_0}{\omega}\right)} \to -\infty$$ 完美电导体（PEC)

- 高频、等离子体，当$\omega > \omega_p$，$\omega \gg \gamma_0$

$$\varepsilon_r = 1 - \frac{\omega_p^2}{\omega^2} \qquad 0 < \varepsilon_r < 1$$

同一材料在不同波段也可能具有迥异的光学性质

导体存在时的电磁波

导体内部存在自由电子，在电磁波的作用下，自由电子运动形成传导电流，产生焦耳热损耗，从而使得导体中的电磁波衰减

1. 导体内的自由电荷分布

设导体内某区域内有自由电荷分布ρ，则有$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon}$

静电场情况：自由电荷分布在导体表面；

迅变场情况，自由电荷如何分布？

在电场作用下，导体内引起传导电流 $\vec{J} = \sigma \vec{E}$

根据流守恒定律 $$\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$$

可以得到：$$\nabla \cdot (\sigma \vec{E}) = \frac{\sigma \rho}{\varepsilon} = -\frac{\partial \rho}{\partial t}$$

$$\rho(t) = \rho_0 e^{-\frac{\sigma}{\varepsilon} t}$$

特征衰减时间$\tau = \frac{\varepsilon}{\sigma}$

当电磁波频率$\omega \ll \tau^{-1}$，$\rho(t)$可以认为约等于0，导体内部无电荷积累

$\omega \ll \tau^{-1}$ ⇒ $\frac{\sigma}{\varepsilon_0 \omega} \gg 1$ 良导体条件

导体存在时的电磁波

2. 导体内部的电磁波

$$\begin{cases} \nabla\cdot\vec{D}=\rho_f \\ \nabla\times\vec{E}=-\dfrac{\partial\vec{B}}{\partial t} \\ \nabla\cdot\vec{B}=0 \\ \nabla\times\vec{H}=\vec{J}+\dfrac{\partial\vec{D}}{\partial t} \end{cases}$$

$$\vec{J}=\sigma\vec{E}$$

时谐场

$$\begin{cases} \nabla\cdot\vec{E}=0 \\ \nabla\times\vec{E}=i\omega\vec{B} \\ \nabla\cdot\vec{B}=0 \\ \nabla\times\vec{H}=\sigma\vec{E}-i\omega\varepsilon_0\vec{E} \end{cases}$$

$$\begin{aligned} \nabla\times\vec{H}&=\sigma\vec{E}-i\omega\varepsilon_0\vec{E} \\ &=-i\omega\varepsilon_0\left(1+\frac{i\sigma}{\omega\varepsilon_0}\right)\vec{E} \\ &=-i\omega\varepsilon_0\left(1+\frac{i\sigma}{\omega\varepsilon_0}\right)\vec{E} \\ &=-i\omega\varepsilon\vec{E} \end{aligned}$$

导体的介电系数：

$$\varepsilon=\varepsilon_0\left(1+\frac{i\sigma}{\omega\varepsilon_0}\right)$$

$$\begin{cases} (\nabla^2+k^2)\vec{E}(\vec{x})=0 \\ \nabla\cdot\vec{E}=0 \\ \vec{B}=\dfrac{1}{i\omega}\nabla\times\vec{E} \end{cases}$$

色散关系：

$$k^2=\omega^2\varepsilon\mu$$

导体存在时的电磁波

导体内部的电磁波满足：

$$\begin{cases}(\nabla^2+k^2)\vec{E}(\vec{x})=0\\ \nabla\cdot\vec{E}=0\\ \vec{B}=\dfrac{1}{i\omega}\nabla\times\vec{E}\end{cases}$$

$$\varepsilon=\varepsilon_0\left(1+\frac{i\sigma}{\omega\varepsilon_0}\right)$$

其中波矢：　$k^2=\omega^2\varepsilon\mu$　复数波矢量

令波矢：$\vec{k}=\vec{\alpha}+i\vec{\beta}$

$$\begin{cases}\alpha^2-\beta^2=\omega^2\varepsilon_0\mu\\ \vec{\alpha}\cdot\vec{\beta}=\omega\mu\sigma\end{cases}$$

相位常数　衰减常数

- 平面波解：　$\vec{E}(\vec{x},t)=\vec{E}_0e^{i(\vec{k}\cdot\vec{x}-\omega t)}=\vec{E}_0e^{i[(\vec{\alpha}+i\vec{\beta})\cdot\vec{x}-\omega t]}$

$$=\vec{E}_0e^{-\vec{\beta}\cdot\vec{x}}e^{i\vec{\alpha}\cdot\vec{x}}e^{-i\omega t}$$

衰减波

- 良导体条件下$\dfrac{\sigma}{\varepsilon_0\omega}\gg1$，$\varepsilon\to\dfrac{i\sigma}{\omega}$

趋肤深度　$\delta=\sqrt{2/(\omega\mu\sigma)}$

复波矢　$k=\omega\sqrt{\mu\dfrac{i\sigma}{\omega}}=\dfrac{1}{\sqrt{2}}(1+i)\sqrt{\omega\mu\sigma}=\dfrac{1+i}{\delta}$

导体存在时的电磁波

设平面电磁波沿着z方向传输

$$\vec{E}(\vec{x},t)=\vec{E}_0e^{-\vec{\beta}\cdot\vec{x}}e^{i\vec{\alpha}\cdot\vec{x}}e^{-i\omega t}$$
$$=\vec{E}_0e^{-z/\delta}e^{i(z/\delta-\omega t)}$$

沿着z方向衰减

$$\frac{\sigma}{\varepsilon_0\omega}\gg 1$$

准静场

- 由$\nabla\cdot\vec{E}=0$可得$\vec{k}\cdot\vec{E}=0$, 即$\hat{z}\cdot\vec{E}_0=0$, 横波性质不变
- 由$\vec{B}=\frac{1}{i\omega}\nabla\times\vec{E}$可得$\vec{B}=\frac{\vec{k}}{\omega}\times\vec{E}$

$$\vec{B}=\frac{k}{\omega}\hat{z}\times\vec{E}=\frac{1+i}{\delta\omega}\hat{z}\times\vec{E}=\sqrt{\frac{\mu\sigma}{\omega}}e^{i\pi/4}\hat{z}\times\vec{E}$$

电磁场不再同相位，磁场落后$i\pi/4$相位

- 磁场能$U_B=\frac{1}{2\mu}B^2=\frac{1}{2}\frac{\sigma}{\omega\varepsilon}\varepsilon E^2=\frac{1}{2}\frac{\sigma}{\omega\varepsilon}U_E\gg U_E$, 磁场能远大于电场能

导体存在时的电磁波

➢ 导体在光波段的响应

光波段，导体的电导率的虚部占主要贡献，其相对介电系数

$$\varepsilon_r = 1 - \frac{\omega_p^2}{\omega^2}$$

其波矢量为：

$$k^2 = \omega^2\varepsilon\mu = \omega^2\varepsilon_0\mu_0\sqrt{\varepsilon_r}$$

$$= \frac{\omega^2}{c^2}\sqrt{1 - \frac{\omega_p^2}{\omega^2}} = \frac{1}{c^2}\sqrt{\omega^2 - \omega_p^2}$$

$$\delta = \frac{c}{\sqrt{\omega_p^2 - \omega^2}}$$

- 当$\omega < \omega_p$时，$k = i/\delta$, 此时的电磁波

$$\vec{E}(\vec{x}, t) = \vec{E}_0 e^{i(\vec{k}\cdot\vec{x} - \omega t)} = \vec{E}_0 e^{-z/\delta} e^{-i\omega t}$$

倏逝波

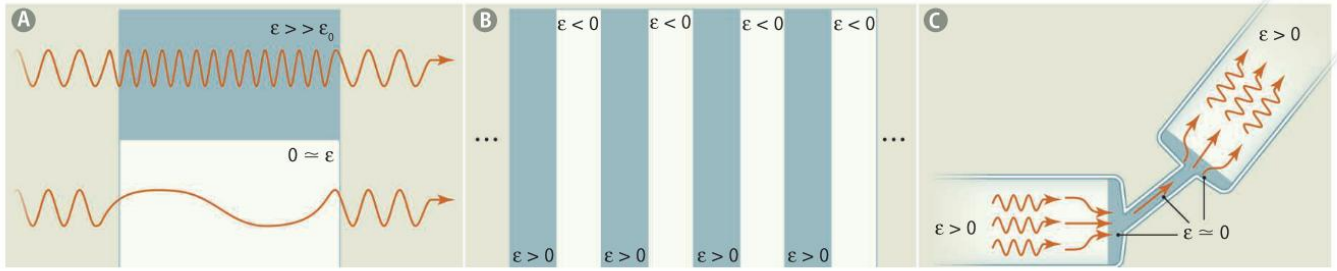
$$\vec{B} = \frac{1}{i\omega}\nabla \times \vec{E} = \frac{k}{\omega}\hat{z} \times \vec{E} = \frac{i}{\delta\omega}\hat{z} \times \vec{E} = \frac{1}{\delta\omega}e^{i\frac{\pi}{2}}\hat{z} \times \vec{E}$$

磁场比电场落后$\frac{\pi}{2}$相位

导体存在时的电磁波

- 当$\omega = \omega_p$时，$\varepsilon_r \to 0$, $\delta \to \infty$, $\left|\vec{B}\right| \to 0$, 纯电场, 近零折射率

$$\delta = \frac{c}{\sqrt{\omega_p^2 - \omega^2}}$$



Features of ENZ metamaterials. (**A**) The wavelength can be "stretched" within materials with low permittivity, whereas for high permittivity it is compressed. (**B**) A judicious mixture of a permittivity-positive and a permittivity-negative constituent structure may provide an effectively ENZ metamaterial. (**C**) Supercoupling phenomenon in ENZ-filled ultranarrow channels between two similar, but arbitrarily oriented, waveguides, in which an efficient tunneling occurs regardless of the length, shape, bending, and twisting of the ultranarrow channel.

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- 当$\omega > \omega_p$时，$0 < \varepsilon_r < 1$, $n = \sqrt{\varepsilon_r}$, 电磁波能穿透导体，紫外透明

$$\varepsilon_r = 1 - \frac{\omega_p^2}{\omega^2}$$

导体存在时的电磁波

- 非良导体，电介质，一方面价带电子贡献ε ，另一方面自由电子或者掺杂电荷贡献σ_c, 但是$\frac{\sigma_c}{\omega\varepsilon_0} \leq 1$，

$$\tilde{\varepsilon}_r(\omega) = \varepsilon_r + \frac{i\sigma_c}{\omega\varepsilon_0} \qquad \text{代入色散关系} \qquad k^2 = \omega^2\varepsilon\mu = \frac{\omega^2}{c^2}\tilde{\varepsilon}_r$$

$$k = \frac{\omega}{c}\sqrt{\tilde{\varepsilon}_r} = \frac{\omega}{c}\sqrt{\varepsilon_r + \frac{i\sigma_c}{\omega\varepsilon_0}} \approx \frac{\omega}{c}\sqrt{\varepsilon_r} + \frac{1}{2}\frac{i\sigma_c}{\omega\varepsilon_0}\frac{1}{\sqrt{\varepsilon_r}}$$
$$= \alpha + i\beta$$

代入平面电磁波解

$$\vec{E}(\vec{x},t) = \vec{E}_0 e^{i[(\alpha+i\beta)z-\omega t]} = \vec{E}_0 e^{-i\beta z} e^{i(\alpha z-\omega t)}$$

- 折射率的虚部代表波传播过程中的损耗，与电导率相反
- 与真空中类似，但是波传播过程中有少量能量损耗

电磁波在各向异性介质中的传播

本构关系 ⇒ 色散关系 ⇒ 本征态（电场、磁场关系，偏振等...)

1. 各项异性介质的本构关系

本构关系 $\vec{D} = \overleftrightarrow{\varepsilon} \cdot \vec{E}$ $D_i = \sum_j \varepsilon_{ij} E_j = \varepsilon_{ij} E_j$

$\overleftrightarrow{\varepsilon}$矩阵对角化 $\overleftrightarrow{\varepsilon} = \begin{pmatrix} \varepsilon_{11} & 0 & 0 \\ 0 & \varepsilon_{22} & 0 \\ 0 & 0 & \varepsilon_{33} \end{pmatrix}$ ε_{11}，ε_{22}，ε_{33}主轴介电系数

$D_i = \varepsilon_{ii} E_i \ (i = 1,2,3)$

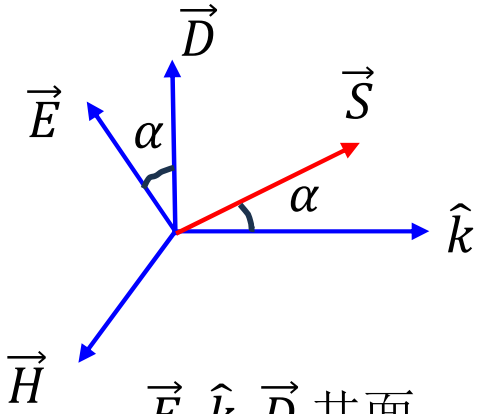
$\varepsilon_{ii} = \varepsilon_0 \varepsilon_{ri} = \varepsilon_0 {n_i}^2$ n_i为折射率

$$\begin{cases} n_x = n_y = n_z \text{ 各项同性，立方晶系} \\ n_x = n_y \neq n_z \text{，单轴晶体，三方、四方、六方晶系} \\ n_x \neq n_y \neq n_z \text{，双轴晶体，三斜、单斜，正交晶系} \end{cases}$$

电磁波在各向异性介质中的传播

2. 色散关系及本征态

考虑非磁$\mu = \mu_0$、无损耗$\sigma = 0$介质中的平面波解



$\vec{E}, \hat{k}, \vec{D}$ 共面

$$(\vec{E}, \vec{D}, \vec{B}, \vec{H}) = (\vec{E}_0, \vec{D}_0, \vec{B}_0, \vec{H}_0) e^{i\vec{k}\cdot\vec{x}} e^{-i\omega t}$$

$$\begin{cases} \nabla \cdot \vec{D} = 0 \\ \nabla \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t} \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{H} = \dfrac{\partial \vec{D}}{\partial t} \end{cases} \Rightarrow \begin{cases} \vec{k} \cdot \vec{D} = 0 \\ \vec{k} \times \vec{E} = \omega \mu_0 \vec{H} \\ \vec{k} \cdot \vec{H} = 0 \\ \vec{k} \times \vec{H} = -\omega \vec{D} \end{cases}$$

$\hat{k} \perp (\vec{H}, \vec{D})$

- 电位移方向与电场方向不一定一致
- 能流方向与波矢方向不一定一致

电磁波在各向异性介质中的传播

$$\vec{k}\times(\vec{k}\times\vec{E})=\vec{k}\times(\omega\mu_0\vec{H})=-\omega^2\mu_0\vec{D}$$

$$\vec{k}\times\vec{H}=-\omega\vec{D}$$

可得：$$\vec{D}=-\frac{1}{\omega^2\mu_0}\vec{k}\times(\vec{k}\times\vec{E})=\frac{k^2}{\omega^2\mu_0}[\vec{E}-\hat{k}(\hat{k}\cdot\vec{E})]$$

电位移矢量与电场的一般关系式

主轴方向上满足 $$D_i=\frac{1}{\omega^2\mu_0}\left[k^2E_i-k_i(\vec{k}\cdot\vec{E})\right]$$

$$D_i=\varepsilon_{ii}E_i\ (i=1,2,3)$$

同时在主轴上 $$D_i=\varepsilon_i E_i$$

$$\Rightarrow\quad \omega^2\mu_0 D_i=k^2\frac{D_i}{\varepsilon_i}-k_i(\vec{k}\cdot\vec{E})$$

$$D_i=\frac{k_i(\vec{k}\cdot\vec{E})}{\frac{k^2}{\varepsilon_i}-\omega^2\mu_0}=\frac{k_i(\vec{k}\cdot\vec{E})}{k^2(\frac{1}{\varepsilon_i}-\frac{\omega^2\mu_0}{k^2})}$$

若取$k=k_0n$, n为电磁波传播的有效折射率，则$k^2=k_0^2n^2$

$$D_i=\frac{k_i(\vec{k}\cdot\vec{E})}{k^2(\frac{1}{\varepsilon_i}-\frac{\omega^2\mu_0}{k^2})}=\frac{k_i(\vec{k}\cdot\vec{E})}{k^2(\frac{1}{\varepsilon_i}-\frac{\mu_0c^2}{n^2})}$$

$$\vec{D}=\frac{n^2}{c^2\mu_0}[\vec{E}-\hat{k}(\hat{k}\cdot\vec{E})]$$

电磁波在各向异性介质中的传播

由公式$\vec{k}\cdot\vec{D}=0$，可得：

$$D_i=\frac{k_i(\vec{k}\cdot\vec{E})}{k^2(\frac{1}{\varepsilon_i}-\frac{\mu_0c^2}{n^2})}$$

$$\frac{k_x^2(\vec{k}\cdot\vec{E})}{k^2(\frac{1}{\varepsilon_x}-\frac{\mu_0c^2}{n^2})}+\frac{k_y^2(\vec{k}\cdot\vec{E})}{k^2(\frac{1}{\varepsilon_y}-\frac{\mu_0c^2}{n^2})}+\frac{k_z^2(\vec{k}\cdot\vec{E})}{k^2(\frac{1}{\varepsilon_z}-\frac{\mu_0c^2}{n^2})}=0$$

$$\frac{k_x^2}{\frac{1}{\varepsilon_x}-\frac{\mu_0c^2}{n^2}}+\frac{k_y^2}{\frac{1}{\varepsilon_y}-\frac{\mu_0c^2}{n^2}}+\frac{k_z^2}{\frac{1}{\varepsilon_z}-\frac{\mu_0c^2}{n^2}}=0$$

Fresnel方程

$$k_x^2+k_y^2+k_z^2=k^2=k_0^2n^2$$

若令$v_i=\frac{1}{\sqrt{\varepsilon_i\mu_0}}$为主轴相速度，待求波相速度$v_p=\frac{c}{n}$，Fresnel方程可以改写为：

$$\frac{k_x^2}{v_p^2-v_x^2}+\frac{k_y^2}{v_p^2-v_y^2}+\frac{k_z^2}{v_p^2-v_z^2}=0$$

或者

$$\frac{k_x^2}{\frac{1}{n_x^2}-\frac{1}{n^2}}+\frac{k_y^2}{\frac{1}{n_y^2}-\frac{1}{n^2}}+\frac{k_z^2}{\frac{1}{n_z^2}-\frac{1}{n^2}}=0$$

ε_x，ε_y，ε_z由介质的性质决定，k或n或v_p待求

电磁波在各向异性介质中的传播

➢ 一般情况下，给定波的传播方向，允许两个特殊的线偏振光传输，即由两组解：

n', n''都满足Fresnel方程，则：

$$D_i = \frac{k_i(\vec{k}\cdot\vec{E})}{k^2(\frac{1}{\varepsilon_i}-\frac{\mu_0 c^2}{n^2})}$$

$$\vec{D}'\cdot\vec{D}'' = \frac{\vec{k}\cdot\vec{E}}{k^2}\left[\frac{k_x^2}{\left(\frac{1}{\varepsilon_x}-\frac{\mu_0c^2}{n'^2}\right)\left(\frac{1}{\varepsilon_x}-\frac{\mu_0c^2}{n''^2}\right)}+\frac{k_y^2}{\left(\frac{1}{\varepsilon_y}-\frac{\mu_0c^2}{n'^2}\right)\left(\frac{1}{\varepsilon_y}-\frac{\mu_0c^2}{n''^2}\right)}+\frac{k_z^2}{\left(\frac{1}{\varepsilon_z}-\frac{\mu_0c^2}{n'^2}\right)\left(\frac{1}{\varepsilon_z}-\frac{\mu_0c^2}{n''^2}\right)}\right]$$

$$= \frac{\vec{k}\cdot\vec{E}}{k^2}\left[\left(\frac{k_x^2}{\frac{1}{\varepsilon_x}-\frac{\mu_0c^2}{n'^2}}-\frac{k_x^2}{\frac{1}{\varepsilon_x}-\frac{\mu_0c^2}{n''^2}}\right)\frac{n''^2n'^2}{\mu_0c^2(n''^2-n'^2)}+\cdots\right]$$

$$= \frac{\vec{k}\cdot\vec{E}}{k^2}\frac{n''^2n'^2}{\mu_0c^2(n''^2-n'^2)}\left[\frac{k_x^2}{\frac{1}{\varepsilon_x}-\frac{\mu_0c^2}{n'^2}}+\frac{k_y^2}{\frac{1}{\varepsilon_y}-\frac{\mu_0c^2}{n'^2}}+\frac{k_z^2}{\frac{1}{\varepsilon_z}-\frac{\mu_0c^2}{n'^2}}-\frac{k_x^2}{\frac{1}{\varepsilon_x}-\frac{\mu_0c^2}{n''^2}}-\cdots\right]$$

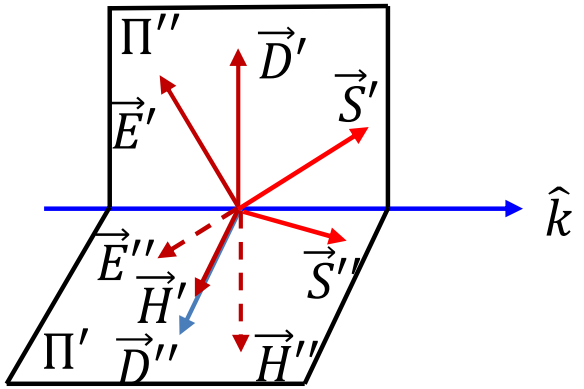
$=0$

该项为0，满足Fresnel方程

两组偏振光的振动面垂直，但是具有不同的有效折射率

电磁波在各向异性介质中的传播

- 两组解，能流在两个面Π', Π''上
- $\vec{E},\vec{k},\vec{D},\vec{S}$ 共面



$\vec{E},\vec{k},\vec{D}$ 共面

➢ 双折射效应，对于单轴晶体　$n_x = n_y \neq n_z$

Z为主轴方向，对于主轴方向，折射率具有旋转对称性　$n_x = n_y = n_o$

$n_z = n_e$

$$v_x = v_y = v_o = \frac{c}{n_o} \qquad v_e = \frac{c}{n_e}$$

满足Fresnel公式　$$\frac{k_x^2}{v_p^2 - v_o^2} + \frac{k_y^2}{v_p^2 - v_o^2} + \frac{k_z^2}{v_p^2 - v_e^2} = 0$$

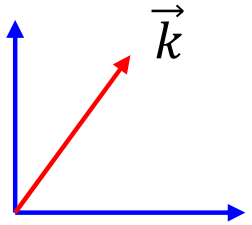
$$\left(v_p^2 - v_o^2\right)\left[\left(v_p^2 - v_e^2\right)\left(k_x^2 + k_y^2\right) + \left(v_p^2 - v_o^2\right)k_z^2\right] = 0$$

电磁波在各向异性介质中的传播

$$\left(v_p^2 - v_o^2\right)\left[\left(v_p^2 - v_e^2\right)\left(k_x^2 + k_y^2\right) + \left(v_p^2 - v_o^2\right)k_z^2\right] = 0$$

- 取电磁波在y-z平面传输，与光轴z方向的夹角为θ，则：$k_x = 0, k_y = k\sin\theta, k_z = k\cos\theta$,上式存在两组解:

$$v_p^2 = v_o^2, \quad v_p^2 = v_o^2\cos^2\theta + v_e^2\sin^2\theta$$



对应的有效折射率 $n' = n_o$

$$n'' = \frac{n_o n_e}{\sqrt{n_e^2\cos^2\theta + n_o^2\sin^2\theta}}$$

因此在单轴晶体中，存在两种不同的光线

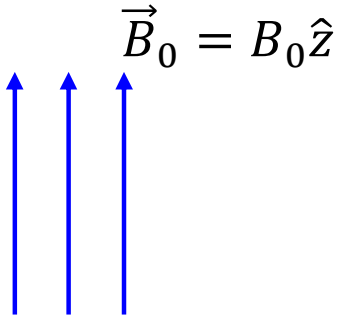
1. 一种$n' = n_o$，不依赖于波矢$\vec{k}$的方向，寻常光；
2. 另一种$n''(\theta)$，依赖于$\vec{k}$的方向，异常光，当$\theta = 0$，$n'' = n_o$，表示沿着光轴方向，无双折射效应；当$\theta = \pi/2$，$n'' = n_e$

旋光介质中的电磁波

旋光介质，考虑磁场导致的旋光效应，磁场中的自由电子气

1. 本构关系

考虑处于静磁场$\vec{B}_0 = B_0\hat{z}$中的自由电子气对电磁场的响应，忽略杂质的散射，静磁场远大于电磁波的磁场



$$m\frac{d\vec{v}}{dt} = -e(\vec{E} + \vec{v}\times\vec{B}_0)$$

时谐场，将试解$\vec{v} = \vec{v}_0 e^{-i\omega t}$代入，并写成分量形式，可得：

$$\begin{cases} -im\omega v_{0x} = -e(E_{0x} + B_0 v_{0y}) \\ -im\omega v_{0y} = -e(E_{0y} - B_0 v_{0x}) \\ -im\omega v_{0z} = -eE_{0z} \end{cases} \Rightarrow \begin{cases} v_{0x} = \dfrac{-e}{m}\dfrac{-i\omega E_{0x} - \omega_B E_{0y}}{\omega_B^2 - \omega^2} \\ v_{0y} = \dfrac{-e}{m}\dfrac{-i\omega E_{0y} + \omega_B E_{0x}}{\omega_B^2 - \omega^2} \\ v_{0z} = \dfrac{eE_{0z}}{im\omega} \end{cases}$$

$\omega_B = \frac{eB_0}{m}$ 旋磁频率

旋光介质中的电磁波

将电子速度表达式代入欧姆定律，可得电导率 $\vec{J}=\overleftrightarrow{\sigma}\cdot\vec{E}=n(-e)\vec{v}$

$$\begin{cases} J_x=\dfrac{ne^2}{m}\dfrac{-i\omega E_{0x}-\omega_B E_{0y}}{\omega_B^2-\omega^2}=\dfrac{\varepsilon_0\omega_p^2(-i\omega E_{0x}-\omega_B E_{0y})}{\omega_B^2-\omega^2} \\ J_y=\dfrac{ne^2}{m}\dfrac{-i\omega E_{0y}+\omega_B E_{0x}}{\omega_B^2-\omega^2}=\dfrac{\varepsilon_0\omega_p^2(-i\omega E_{0y}+\omega_B E_{0x})}{\omega_B^2-\omega^2} \\ J_z=\dfrac{ne^2}{m}\dfrac{iE_{0z}}{\omega}=\dfrac{i\varepsilon_0\omega_p^2 E_{0z}}{\omega} \end{cases}$$

$$\omega_p=\sqrt{\frac{ne^2}{m\varepsilon_0}}$$

等离子频率

由 $\vec{J}=\overleftrightarrow{\sigma}\cdot\vec{E}$，可以得电导率张量

$$\overleftrightarrow{\sigma}=\frac{\varepsilon_0\omega_p^2}{\omega^2-\omega_B^2}\begin{pmatrix} i\omega & \omega_B & 0 \\ -\omega_B & i\omega & 0 \\ 0 & 0 & i\dfrac{\omega^2-\omega_B^2}{\omega} \end{pmatrix}$$

等效介电系数张量由 $\overleftrightarrow{\varepsilon}_r=\overleftrightarrow{I}+\frac{i\overleftrightarrow{\sigma}}{\omega\varepsilon_0}$

$$\overleftrightarrow{\varepsilon}_r=\overleftrightarrow{I}+\frac{i\omega_p^2}{\omega(\omega^2-\omega_B^2)}\begin{pmatrix} i\omega & \omega_B & 0 \\ -\omega_B & i\omega & 0 \\ 0 & 0 & i\dfrac{\omega^2-\omega_B^2}{\omega} \end{pmatrix}$$

旋光介质中的电磁波

$$\overleftrightarrow{\varepsilon}_r = \overleftrightarrow{I} + \frac{i\omega_p^2}{\omega(\omega^2-\omega_B^2)}\begin{pmatrix} i\omega & \omega_B & 0 \\ -\omega_B & i\omega & 0 \\ 0 & 0 & i\dfrac{\omega^2-\omega_B^2}{\omega} \end{pmatrix}$$

$$\varepsilon_{r11} = \varepsilon_1 = 1 - \frac{\omega_p^2}{\omega^2-\omega_B^2}$$

$$\varepsilon_{r12} = i\varepsilon_2 = \frac{i\omega_p^2\omega_B}{\omega(\omega^2-\omega_B^2)}$$

$$\varepsilon_{r33} = \varepsilon_3 = 1 - \frac{\omega_p^2}{\omega^2}$$

$$\overleftrightarrow{\varepsilon}_r = \begin{pmatrix} \varepsilon_1 & i\varepsilon_2 & 0 \\ -i\varepsilon_2 & \varepsilon_1 & 0 \\ 0 & 0 & \varepsilon_3 \end{pmatrix}$$

磁场对于自由电子气的影响：

（1）各向异性得电磁响应；

（2）存在虚得非对角元；

- 当$B_0 \to 0, \omega_B \to 0$, 介电系数回到Drude模型

$$\varepsilon_r = 1 - \frac{\omega_p^2}{\omega^2}$$

旋光介质中的电磁波

2. 旋光介质中得电磁波本征态

考虑沿着z方向传播得平面波解

$$(\vec{E},\vec{H}) = (\vec{E}_0,\vec{H}_0)e^{i\vec{k}\cdot\vec{x}}e^{-i\omega t}$$

$$\vec{D} = \varepsilon_0\overleftrightarrow{\varepsilon}_r\cdot\vec{E}$$

$$\begin{cases} \nabla\cdot\vec{D} = 0 \\ \nabla\times\vec{E} = -\dfrac{\partial\vec{B}}{\partial t} \\ \nabla\cdot\vec{B} = 0 \\ \nabla\times\vec{H} = \dfrac{\partial\vec{D}}{\partial t} \end{cases} \Rightarrow \begin{cases} \vec{k}\cdot(\overleftrightarrow{\varepsilon}_r\cdot\vec{E}_0) = 0 \\ \vec{k}\times\vec{E}_0 = \omega\mu_0\vec{H}_0 \\ \vec{k}\cdot\vec{H}_0 = 0 \\ \vec{k}\times\vec{H}_0 = -\omega\varepsilon_0\overleftrightarrow{\varepsilon}_r\cdot\vec{E}_0 \end{cases}$$

假设：（1）$\omega > \omega_p \gg \omega_B$，则有$\varepsilon_1$，$\varepsilon_2$正实数，且$\varepsilon_1 \gg \varepsilon_2$；（2）电磁波沿着磁场方向，即z传播 $\vec{k} = k\hat{z}$

旋光介质中的电磁波

由第一式 $\vec{k}\cdot(\overleftrightarrow{\varepsilon}_r\cdot\vec{E}_0)=0$

$$\overleftrightarrow{\varepsilon}_r=\begin{pmatrix}\varepsilon_1 & i\varepsilon_2 & 0\\ -i\varepsilon_2 & \varepsilon_1 & 0\\ 0 & 0 & \varepsilon_3\end{pmatrix}$$

可得： $\hat{z}\cdot\vec{E}_0=0$，即$E_{0z}=0$, 此时得电磁波仍然为横波

$$\vec{k}=k\hat{z}$$

由第二式 $\vec{k}\times\vec{E}_0=\omega\mu_0\vec{H}_0$

$$\vec{k}\times\vec{H}_0=-\omega\varepsilon_0\overleftrightarrow{\varepsilon}_r\cdot\vec{E}$$

$$\vec{k}\times(\vec{k}\times\vec{E}_0)=\vec{k}(\vec{k}\cdot\vec{E}_0)-k^2\vec{E}_0=\omega\mu_0\vec{k}\times\vec{H}_0$$

$$=\omega\mu_0(-\omega\varepsilon_0\overleftrightarrow{\varepsilon}_r\cdot\vec{E})=-k_0^2\overleftrightarrow{\varepsilon}_r\cdot\vec{E}_0$$

得到关于电场得本征值方程： $k^2\vec{E}_0-k_0^2\overleftrightarrow{\varepsilon}_r\cdot\vec{E}_0=0$

$$\begin{pmatrix}k^2-k_0^2\varepsilon_1 & -ik_0^2\varepsilon_2\\ ik_0^2\varepsilon_2 & k^2-k_0^2\varepsilon_1\end{pmatrix}\begin{pmatrix}E_{0x}\\ E_{0y}\end{pmatrix}=0$$

其本征值为 $\begin{vmatrix}k^2-k_0^2\varepsilon_1 & -ik_0^2\varepsilon_2\\ ik_0^2\varepsilon_2 & k^2-k_0^2\varepsilon_1\end{vmatrix}=0$ ⇒ $\begin{cases}k_+=k_0\sqrt{\varepsilon_1+\varepsilon_2}\\ k_-=k_0\sqrt{\varepsilon_1-\varepsilon_2}\end{cases}$

旋光介质中的电磁波

$$\begin{cases} k_+ = k_0\sqrt{\varepsilon_1 + \varepsilon_2} \\ k_- = k_0\sqrt{\varepsilon_1 - \varepsilon_2} \end{cases}$$

代入本征方程：

$$\begin{cases} E_{0x} = iE_{0y} \ (k = k_+) \\ E_{0x} = -iE_{0y} \ (k = k_-) \end{cases}$$

$$\begin{cases} k_+ = k_0\sqrt{\varepsilon_1 + \varepsilon_2} \\ k_- = k_0\sqrt{\varepsilon_1 - \varepsilon_2} \end{cases}$$

可以得到相应的本征态

$$\begin{cases} E_0^+ = E_0\hat{x} + \dfrac{E_0\hat{y}}{i} = \sqrt{2}E_0\dfrac{\hat{x} - i\hat{y}}{\sqrt{2}} \ (k = k_+) & \text{右旋偏振光} \\ E_0^- = E_0\hat{x} + iE_0\hat{y} = \sqrt{2}E_0\dfrac{\hat{x} + i\hat{y}}{\sqrt{2}} \ (k = k_+) & \text{左旋偏振光} \end{cases}$$

电磁波的本征态为沿z传播的圆偏振电磁波，其相速度分别为：

$$\begin{cases} v_R = \dfrac{\omega}{k_+} = \dfrac{c}{\sqrt{\varepsilon_1 + \varepsilon_2}} & \text{(右旋偏振)} \\ v_L = \dfrac{\omega}{k_-} = \dfrac{c}{\sqrt{\varepsilon_1 - \varepsilon_2}} & \text{(左旋偏振)} \end{cases}$$

- 左旋偏振光的相速度超过右旋偏振光 $v_R < v_L$

旋光介质中的电磁波

3. 法拉第效应（磁致旋光）

线偏振光进入介质中，由于左、右旋偏振光波速不同，电磁波偏振面以前进方向为轴旋转

设进入法拉第介质之前线偏振的电磁波为 $E_0\hat{x}e^{i(k_0z-\omega t)}$,可以分解为圆偏振电磁波的叠加

$$E_0\hat{x}e^{i(k_0z-\omega t)}=\frac{E_0}{\sqrt{2}}(\hat{e}_R+\hat{e}_L)e^{i(k_0z-\omega t)}$$

$$\begin{cases}\hat{e}_R=\dfrac{\hat{x}-i\hat{y}}{\sqrt{2}}\\ \hat{e}_L=\dfrac{\hat{x}+i\hat{y}}{\sqrt{2}}\end{cases}$$

进入法拉第介质之后，假设左右旋电磁波透射率相同，电磁波为：

$$\frac{E_0}{\sqrt{2}}\left(\hat{e}_Re^{ik_+z}+\hat{e}_Le^{ik_-z}\right)e^{-i\omega t}=\frac{E_0}{\sqrt{2}}(\hat{e}_Re^{i\frac{\Delta kz}{2}}+\hat{e}_Le^{-i\frac{\Delta kz}{2}})e^{i\bar{k}z}e^{-i\omega t}$$

$$=\frac{E_0}{\sqrt{2}}(\hat{e}_Re^{i\frac{\Delta kz}{2}}+\hat{e}_Le^{-i\frac{\Delta kz}{2}})e^{i\bar{k}z}e^{-i\omega t}$$

$$=\frac{E_0}{2}[\hat{x}\left(e^{i\frac{\Delta kz}{2}}+e^{-i\frac{\Delta kz}{2}}\right)-i\hat{y}(e^{i\frac{\Delta kz}{2}}-e^{-i\frac{\Delta kz}{2}})]e^{i\bar{k}z}e^{-i\omega t}$$

$$\bar{k}=\frac{k_++k_-}{2}$$

$$\Delta k=k_+-k_-$$

旋光介质中的电磁波

$$\frac{E_0}{\sqrt{2}}\left(\hat{e}_R e^{ik_+ z}+\hat{e}_L e^{ik_- z}\right)e^{-i\omega t}=\frac{E_0}{2}e^{i\bar{k}z}\left[\hat{x}\left(e^{i\frac{\Delta kz}{2}}+e^{-i\frac{\Delta kz}{2}}\right)-i\hat{y}\left(e^{i\frac{\Delta kz}{2}}-e^{-i\frac{\Delta kz}{2}}\right)\right]e^{-i\omega t}$$

$$=E_0\left(\hat{x}\cos\frac{\Delta kz}{2}+\hat{y}\sin\frac{\Delta kz}{2}\right)e^{i\bar{k}z}e^{-i\omega t}$$

传播过程中电矢量发生旋转

$$\begin{cases}k_+=k_0\sqrt{\varepsilon_1+\varepsilon_2}\\ k_-=k_0\sqrt{\varepsilon_1-\varepsilon_2}\end{cases}$$

设介质的厚度d，出射的偏振为：$\hat{x}\cos\frac{\Delta kd}{2}+\hat{y}\sin\frac{\Delta kd}{2}$

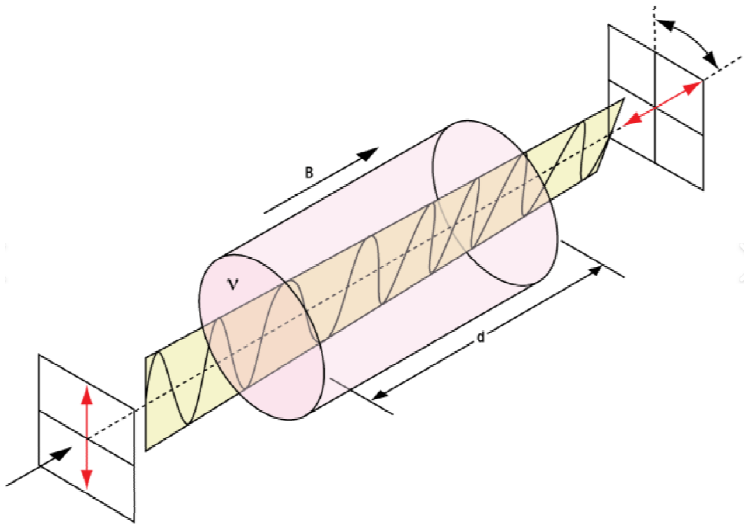
偏振角度为：$\Delta\phi=\arctan\frac{|E_y|}{|E_x|}=\frac{\Delta kd}{2}$

$$=\frac{k_+-k_-}{2}d=\frac{k_0 d}{2}\left(\sqrt{\varepsilon_1+\varepsilon_2}-\sqrt{\varepsilon_1-\varepsilon_2}\right)$$

$$\approx\frac{k_0 d}{2}\sqrt{\varepsilon_1}\frac{\varepsilon_2}{\varepsilon_1}=\frac{k_0 d}{2}\frac{\varepsilon_2}{\sqrt{\varepsilon_1}}$$

取$\varepsilon_2\ll\varepsilon_1$

- 正比于厚度d
- 正比于磁场强度$\varepsilon_2\propto|\vec{B}|$



电磁波在界面的反射与折射

非均匀介质，两介质界面电磁波传播的问题。

1. 电磁波边界条件

$$\hat{n}\cdot\left(\vec{D}_2-\vec{D}_1\right)=\sigma_f \qquad \hat{n}\cdot\left(\vec{B}_2-\vec{B}_1\right)=0$$

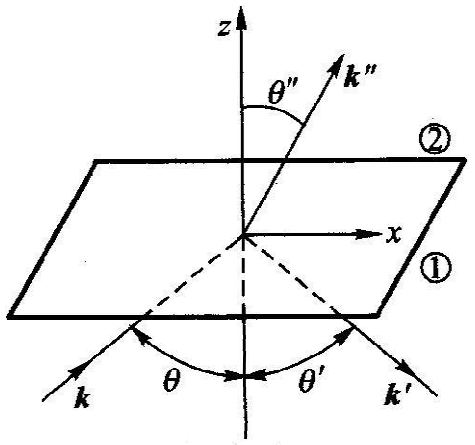
$$\hat{n}\times\left(\vec{E}_2-\vec{E}_1\right)=0 \qquad \hat{n}\times\left(\vec{H}_2-\vec{H}_1\right)=\vec{\alpha}_f$$

对于绝缘介质，$\sigma_f=0$，$\vec{\alpha}_f=0$，高频下的导体自由部分可以归结于复折射率，同样满足该条件。

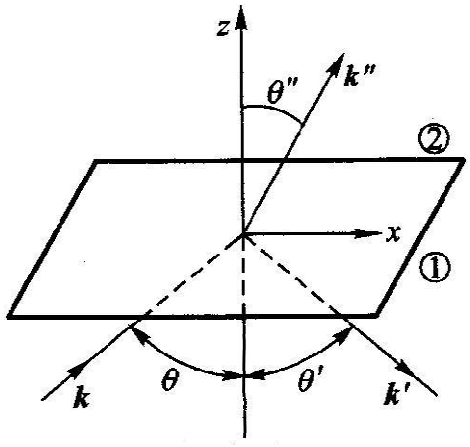
2. 反射、折射定律

电磁波在两介质界面会发生折射、透射，因此考虑入射波、反射波及折射波

$$\begin{cases}\vec{E}=\vec{E}_0e^{i(\vec{k}\cdot\vec{x}-\omega t)} & \text{入射波}\\ \vec{E}'=\vec{E}_0'e^{i(\vec{k}'\cdot\vec{x}-\omega' t)} & \text{反射波}\\ \vec{E}''=\vec{E}_0''e^{i(\vec{k}''\cdot\vec{x}-\omega'' t)} & \text{折射波}\end{cases}$$



电磁波在界面的反射与折射



- 先考虑，反射和折射波的来源是介质的受迫振动，其频率与入射波应该一致，因此有$\omega = \omega' = \omega''$
- 再考虑，反射、折射问题在界面的空间平移不变性，关于空间的相位因子不依赖具体$\vec{x}$，即$e^{i\vec{k}\cdot\vec{x}} = e^{i\vec{k}'\cdot\vec{x}} = e^{i\vec{k}''\cdot\vec{x}}$，即入射、反射及折射波波矢在界面上的切向分量相等，即$\vec{k}_{\parallel} = \vec{k}'_{\parallel} = \vec{k}''_{\parallel}$;
- 再次考虑波矢量的法向分量，考虑因果律，反射、折射波能量离开界面，因此$k'_z = -\sqrt{k^2 - k_{\parallel}^2}, k''_z = -\sqrt{k''^2 - k_{\parallel}^2}$

反射、折射的基本规律：

(1) 频率不变$\omega = \omega' = \omega''$

(2) 切向波矢量相等，若$k_y = 0$, 则$k'_y = k''_y = 0$, 入射、反射及折射波在同一平面内，因此可以定义入射面，即入射波波矢与界面法线构成的平面

电磁波在界面的反射与折射

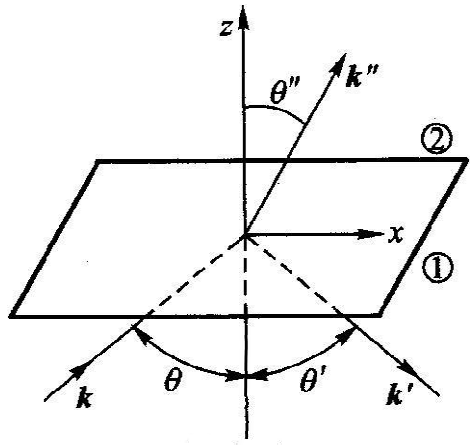
（3） k_z'， k_z''的符号选择由因果律决定

（4） 由$k_x = k_x'$，即$k\sin\theta = k\sin\theta'$, 即 $\theta' = \theta$，满足镜面反射

（5）由$k_x = k_x''$，即$k\sin\theta = k''\sin\theta''$, 即 $\frac{\sin\theta''}{\sin\theta} = \frac{k}{k''} = \frac{\omega\sqrt{\varepsilon\mu}}{\omega\sqrt{\varepsilon''\mu''}}$

$$\frac{\sin\theta''}{\sin\theta} = \frac{\sqrt{\varepsilon\mu}}{\sqrt{\varepsilon''\mu''}} = \frac{n_1}{n_2}$$

Snell定律（1621）

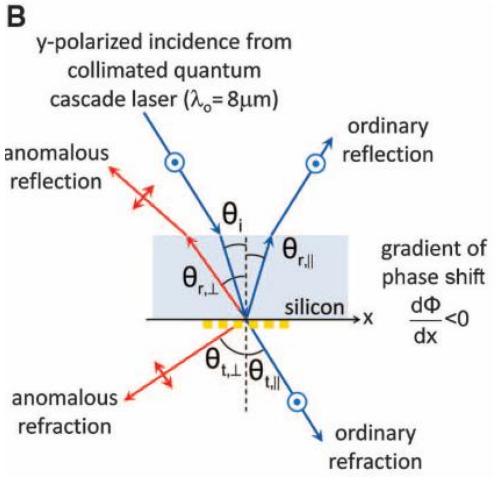


非磁介质$\mu = \mu'' = \mu_0$

Light Propagation with Phase Discontinuities: Generalized Laws of Reflection and Refraction

Nanfang Yu,[1] Patrice Genevet,[1,2] Mikhail A. Kats,[1] Francesco Aieta,[1,3] Jean-Philippe Tetienne,[1,4] Federico Capasso,[1]* Zeno Gaburro[1,5]*

Conventional optical components rely on gradual phase shifts accumulated during light propagation to shape light beams. New degrees of freedom are attained by introducing abrupt phase changes over the scale of the wavelength. A two-dimensional array of optical resonators with spatially varying phase response and subwavelength separation can imprint such phase discontinuities on propagating light as it traverses the interface between two media. Anomalous reflection and refraction phenomena are observed in this regime in optically thin arrays of metallic antennas on silicon with a linear phase variation along the interface, which are in excellent agreement with generalized laws derived from Fermat's principle. Phase discontinuities provide great flexibility in the design of light beams, as illustrated by the generation of optical vortices through use of planar designer metallic interfaces.



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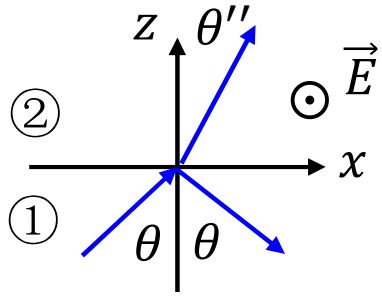
电磁波在界面的反射与折射

3. 振幅关系、菲涅尔公式

由于偏振在垂直于波矢量面内的任意性，相对于入射面，其具有垂直于入射面及平行入射面的两分量，因此可以分开考虑：

（1）TE波，或者s波，电场垂直于入射面

（2）TM波，或者p波，磁场垂直于入射面，电场在入射面内



TE波

(1) 考虑TE波，s波，

$$\begin{cases} \vec{E}_i = \hat{y}Ee^{i(k_x x+k_z z-\omega t)} & \text{入射波} \\ \vec{E}_r = \hat{y}E'e^{i(k_x x-k_z z-\omega t)} & \text{反射波} \\ \vec{E}_t = \hat{y}E''e^{i(k_x x+k_z'' z-\omega t)} & \text{折射波} \end{cases}$$

$$k_z' = -\sqrt{k^2 - k_{\parallel}^2},\ k_z'' = \sqrt{k''^2 - k_{\parallel}^2}$$

由$\vec{H} = \frac{1}{Z}\frac{\vec{k}}{k} \times \vec{E}$，可得磁场

$$Z = \sqrt{\frac{\mu}{\varepsilon}}$$

$$\begin{cases} \vec{H}_i = \dfrac{E}{Z_1}\dfrac{k_x\hat{z} - k_z\hat{x}}{k}e^{i(k_x x+k_z z-\omega t)} & \text{入射波} \\ \vec{H}_r = \dfrac{E'}{Z_1}\dfrac{k_x\hat{z} + k_z\hat{x}}{k}e^{i(k_x x-k_z z-\omega t)} & \text{反射波} \\ \vec{H}_t = \dfrac{E''}{Z_2}\dfrac{k_x\hat{z} - k_z''\hat{x}}{k''}e^{i(k_x x+k_z'' z-\omega t)} & \text{折射波} \end{cases}$$

电磁波在界面的反射与折射

由边值关系在$z=0$,满足：

$$\hat{n}\times(\vec{E}_2-\vec{E}_1)=0$$

$$\hat{n}\times(\vec{H}_2-\vec{H}_1)=0$$

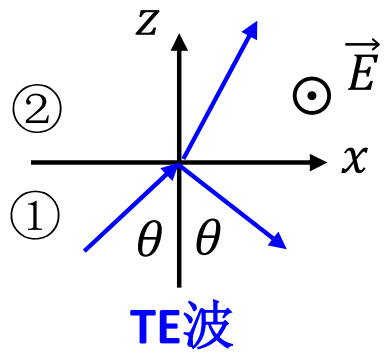
可得：

$$\begin{cases} E+E'=E'' \\ -\dfrac{E}{Z_1}\dfrac{k_z}{k}+\dfrac{E'}{Z_1}\dfrac{k_z}{k}=-\dfrac{E''}{Z_2}\dfrac{k_z''}{k''} \end{cases} \Rightarrow \begin{cases} E'=\dfrac{Z_2\dfrac{k_z}{k}-Z_1\dfrac{k_z''}{k''}}{Z_2\dfrac{k_z}{k}+Z_1\dfrac{k_z''}{k''}}E \\ E''=\dfrac{2Z_2\dfrac{k_z}{k}}{Z_2\dfrac{k_z}{k}+Z_1\dfrac{k_z''}{k''}}E \end{cases}$$

由几何关系 $\dfrac{k_z}{k}=\cos\theta$ $\dfrac{k_z''}{k''}=\cos\theta''$

$$\begin{cases} E'=\dfrac{Z_2\cos\theta-Z_1\cos\theta''}{Z_2\cos\theta+Z_1\cos\theta''}E \\ E''=\dfrac{2Z_2\cos\theta}{Z_2\cos\theta+Z_1\cos\theta''}E \end{cases}$$

阻抗： $Z_i=\sqrt{\dfrac{\mu_i}{\varepsilon_i}}$



阻抗决定了电磁波反射、透射的振幅关系

电磁波在界面的反射与折射

$$\begin{cases} E' = \dfrac{Z_2\cos\theta - Z_1\cos\theta''}{Z_2\cos\theta + Z_1\cos\theta''}E \\ E'' = \dfrac{2Z_2\cos\theta}{Z_2\cos\theta + Z_1\cos\theta''}E \end{cases}$$

TE波，s波

- 若考虑非磁介质 $Z = \sqrt{\dfrac{\mu}{\varepsilon}} = \sqrt{\dfrac{\mu_0}{\varepsilon_0\varepsilon_r}} = \dfrac{Z_0}{\sqrt{\varepsilon_r}} = \dfrac{Z_0}{n}$ 并利用折射定律 $\dfrac{\sin\theta}{\sin\theta''} = \dfrac{n_2}{n_1}$

$$\begin{cases} \dfrac{E'}{E} = \dfrac{n_1\cos\theta - n_2\cos\theta''}{n_1\cos\theta + n_2\cos\theta''} = \dfrac{\sin\theta''\cos\theta - \sin\theta\cos\theta''}{\sin\theta''\cos\theta + \sin\theta\cos\theta''} = -\dfrac{\sin(\theta-\theta'')}{\sin(\theta+\theta'')} \\ \dfrac{E''}{E} = \dfrac{2n_1\cos\theta}{n_1\cos\theta + n_2\cos\theta''} = \dfrac{2\sin\theta''\cos\theta}{\sin\theta''\cos\theta + \sin\theta\cos\theta''} = \dfrac{2\sin\theta''\cos\theta}{\sin(\theta+\theta'')} \end{cases}$$

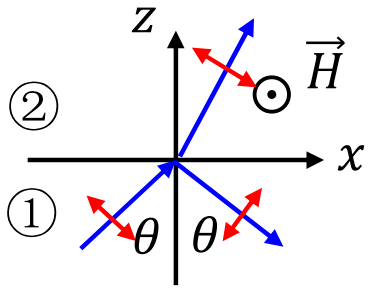
- 当实数的折射角无法定义时，θ''失去几何意义，则$\cos\theta \to \dfrac{k_z}{k}, \cos\theta'' \to \dfrac{k_z''}{k''}$
- 对于TM波（p波），只需做如下代换 $E \to H,\ Z \to \dfrac{1}{Z}$

电磁波在界面的反射与折射

(2) 考虑TM波，p波，

$$\begin{cases} \vec{H}_i = \hat{y}He^{i(k_x x+k_z z-\omega t)} & \text{入射波} \\ \vec{H}_r = \hat{y}H'e^{i(k_x x-k_z z-\omega t)} & \text{反射波} \\ \vec{H}_t = \hat{y}H''e^{i(k_x x+k_z'' z-\omega t)} & \text{折射波} \end{cases}$$

由$\vec{E} = -Z\frac{\vec{k}}{k}\times\vec{H}$，可得电场



TM波

$$\begin{cases} \vec{E}_i = -Z_1 H\dfrac{k_x\hat{z}-k_z\hat{x}}{k}e^{i(k_x x+k_z z-\omega t)} & \text{入射波} \\ \vec{E}_r = -Z_1 H'\dfrac{k_x\hat{z}+k_z\hat{x}}{k}e^{i(k_x x-k_z z-\omega t)} & \text{反射波} \\ \vec{E}_t = -Z_2 H''\dfrac{k_x\hat{z}-k_z''\hat{x}}{k''}e^{i(k_x x+k_z'' z-\omega t)} & \text{折射波} \end{cases}$$

$$E \to H,\ Z \to \frac{1}{Z}$$

由边值关系在$z=0$, 满足：

$$\hat{n}\times(\vec{E}_2-\vec{E}_1)=0$$

$$\hat{n}\times(\vec{H}_2-\vec{H}_1)=0$$

$$\begin{cases} H+H'=H'' \\ -Z_1 H\dfrac{k_z}{k}+Z_1 H'\dfrac{k_z}{k}=-Z_2 H''\dfrac{k_z''}{k''} \end{cases}$$

电磁波在界面的反射与折射

$$\begin{cases} E' = \dfrac{Z_2\dfrac{k_z}{k} - Z_1\dfrac{k_z''}{k''}}{Z_2\dfrac{k_z}{k} + Z_1\dfrac{k_z''}{k''}}E \\ E'' = \dfrac{2Z_2\dfrac{k_z}{k}}{Z_2\dfrac{k_z}{k} + Z_1\dfrac{k_z''}{k''}}E \end{cases}$$

TE波

⇒

做代换$E \to H,\ Z \to \frac{1}{Z}$

TM波

$$\begin{cases} H' = \dfrac{\dfrac{1}{Z_2}\dfrac{k_z}{k} - \dfrac{1}{Z_1}\dfrac{k_z''}{k''}}{\dfrac{1}{Z_2}\dfrac{k_z}{k} + \dfrac{1}{Z_1}\dfrac{k_z''}{k''}}H \\ H'' = \dfrac{2\dfrac{1}{Z_2}\dfrac{k_z}{k}}{\dfrac{1}{Z_2}\dfrac{k_z}{k} + \dfrac{1}{Z_1}\dfrac{k_z''}{k''}}H \end{cases}$$

做代换 $\dfrac{k_z}{k} = \cos\theta$ $\dfrac{k_z''}{k''} = \cos\theta''$

$$\begin{cases} \dfrac{H'}{H} = \dfrac{Z_1\cos\theta - Z_2\cos\theta''}{Z_1\cos\theta + Z_2\cos\theta''} \\ \dfrac{H''}{H} = \dfrac{2Z_1\cos\theta}{Z_1\cos\theta + Z_2\cos\theta''} \end{cases}$$

$Z = \dfrac{Z_0}{n}, \dfrac{\sin\theta}{\sin\theta''} = \dfrac{n_2}{n_1}$

⇒

$$\begin{cases} \dfrac{H'}{H} = \dfrac{n_2\cos\theta - n_1\cos\theta''}{n_2\cos\theta + n_1\cos\theta''} \\ \dfrac{H''}{H} = \dfrac{2n_2\cos\theta}{n_2\cos\theta + n_1\cos\theta''} \end{cases}$$

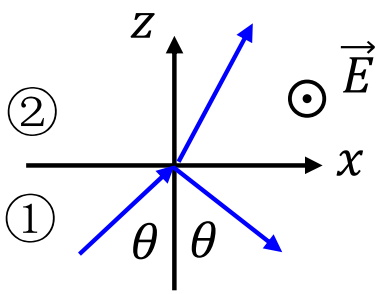
电磁波在界面的反射与折射

$$\begin{cases} \dfrac{H'}{H} = \dfrac{\sin\theta\cos\theta - \sin\theta''\cos\theta''}{\sin\theta\cos\theta + \sin\theta''\cos\theta''} = \dfrac{\tan(\theta-\theta'')}{\tan(\theta+\theta'')} \\ \dfrac{H''}{H} = \dfrac{2\sin\theta\cos\theta}{\sin\theta\cos\theta + \sin\theta''\cos\theta''} = \dfrac{2\sin\theta\cos\theta}{\sin(\theta+\theta'')\cos(\theta-\theta'')} \end{cases}$$

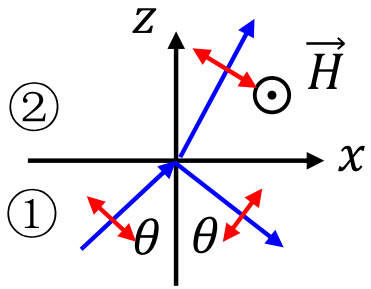
对应的电场的比值为

$$\begin{cases} \dfrac{E'}{E} = \dfrac{Z_1 H'}{Z_1 H} = \dfrac{\tan(\theta-\theta'')}{\tan(\theta+\theta'')} \\ \dfrac{E''}{E} = \dfrac{Z_2 H''}{Z_1 H} = \dfrac{n_1}{n_2}\dfrac{2\sin\theta\cos\theta}{\sin(\theta+\theta'')\cos(\theta-\theta'')} = \dfrac{2\sin\theta''\cos\theta}{\sin(\theta+\theta'')\cos(\theta-\theta'')} \end{cases}$$

电磁波在界面的反射与折射



TE波



TM波

$$\begin{cases} \dfrac{E'}{E} = -\dfrac{\sin(\theta-\theta'')}{\sin(\theta+\theta'')} \\ \dfrac{E''}{E} = \dfrac{2\sin\theta''\cos\theta}{\sin(\theta+\theta'')} \end{cases}$$

$$\begin{cases} \dfrac{E'}{E} = \dfrac{\tan(\theta-\theta'')}{\tan(\theta+\theta'')} \\ \dfrac{E''}{E} = \dfrac{2\sin\theta''\cos\theta}{\sin(\theta+\theta'')\cos(\theta-\theta'')} \end{cases}$$

- 反射率、透射率

反射率：反射波能流与入射波能流的法向分量之比

电磁波在界面的反射与折射

$$R=\frac{\langle\vec{S}_r\rangle\cdot\hat{z}}{\langle\vec{S}_i\rangle\cdot\hat{z}}=\begin{cases}\dfrac{\frac{1}{Z_1}|E'|^2\cos\theta'}{\frac{1}{Z_1}|E|^2\cos\theta}=\left|\dfrac{E'}{E}\right|^2 & \text{TE波 }(s\text{波})\\ \dfrac{Z_1|H'|^2\cos\theta'}{Z_1|H|^2\cos\theta}=\left|\dfrac{H'}{H}\right|^2 & \text{TM波 }(p\text{波})\end{cases}$$

$$R_s=\left|\frac{Z_2\cos\theta-Z_1\cos\theta''}{Z_2\cos\theta+Z_1\cos\theta''}\right|^2 \xrightarrow{Z\to\frac{1}{Z}} R_p=\left|\frac{Z_1\cos\theta-Z_2\cos\theta''}{Z_1\cos\theta+Z_2\cos\theta''}\right|^2$$

透射率：折射波能流与入射波能流的法向分量之比

$$T=\frac{\langle\vec{S}_t\rangle\cdot\hat{z}}{\langle\vec{S}_i\rangle\cdot\hat{z}}=\begin{cases}\dfrac{\frac{1}{Z_2}|E''|^2\cos\theta''}{\frac{1}{Z_1}|E|^2\cos\theta}=\dfrac{Z_1\cos\theta''}{Z_2\cos\theta}\left|\dfrac{E''}{E}\right|^2 & \text{TE波 }(s\text{波})\\ \dfrac{Z_2|H''|^2\cos\theta''}{Z_1|H|^2\cos\theta}=\dfrac{Z_2\cos\theta''}{Z_1\cos\theta}\left|\dfrac{H''}{H}\right|^2 & \text{TM波 }(p\text{波})\end{cases}$$

电磁波在界面的反射与折射

$$T_s = \frac{Z_1 \cos\theta''}{Z_2 \cos\theta}\left|\frac{E''}{E}\right|^2 = \frac{Z_1 \cos\theta''}{Z_2 \cos\theta}\left|\frac{2Z_2 \cos\theta}{Z_2 \cos\theta + Z_1 \cos\theta''}\right|^2$$

$$T_p = \frac{Z_1 \cos\theta''}{Z_2 \cos\theta}\left|\frac{E''}{E}\right|^2 = \frac{Z_2 \cos\theta''}{Z_1 \cos\theta}\left|\frac{2Z_1 \cos\theta}{Z_1 \cos\theta + Z_2 \cos\theta''}\right|^2$$

- 正入射情况$\theta = \theta''$

$$R_s = R_p = \left|\frac{Z_2 - Z_1}{Z_2 + Z_1}\right|^2 \qquad \text{非磁介质} Z = \frac{Z_0}{n} \qquad R_s = R_p = \left|\frac{n_2 - n_1}{n_2 + n_1}\right|^2$$

- 正入射情况，全反射

$$r_s = \frac{Z_2 - Z_1}{Z_2 + Z_1} = \frac{Z_2/Z_1 - 1}{Z_2/Z_1 + 1} = \frac{Z_r - 1}{Z_r + 1} \qquad Z_1 = \sqrt{\frac{\mu_0}{\varepsilon_0}} \qquad Z_r = \sqrt{\frac{\mu_r}{\varepsilon_r}}$$

$|r_s| \to 1$的条件：$r_s = \begin{cases} -1, & \text{即} Z_r \to 0, \varepsilon_r \to \infty \quad \text{完美电导体} \\ 1, & \text{即} Z_r \to \infty, \ \mu_r \to \infty, \quad \text{完美磁导体} \end{cases}$

电磁波在界面的反射与折射

- Brewster角

对于非磁材料 $R_s = \left|\frac{\sin(\theta-\theta'')}{\sin(\theta+\theta'')}\right|^2$ $R_p = \left|\frac{\tan(\theta-\theta'')}{\tan(\theta+\theta'')}\right|^2$

对于TM波（p波），当$\theta+\theta''=\pi/2$，$\tan(\theta+\theta'')\to\infty, R_p\to 0$，该角度下电磁波无反射

$$\theta+\theta''=\frac{\pi}{2} \qquad \frac{\sin\theta}{\sin\theta''}=\frac{\sin\theta}{\sin(\frac{\pi}{2}-\theta)}=\tan\theta=\frac{n_2}{n_1}$$

Brewster角: $\theta=\arctan\frac{n_2}{n_1}$

- 介质全反射

金属表面能实现全反射，但是不适用紫外波段，如何利用介质实现全反射？由折射定律$\frac{\sin\theta}{\sin\theta''}=n_{21}$

电磁波在界面的反射与折射

当$n_1 > n_2$，$n_{21} < 1$, 因此入射角θ小于折射角θ'', 当$\theta'' = \pi/2$，此时的入射角为：

$$\theta_c = \arcsin\frac{n_2}{n_1}$$

当$\theta > \theta_c$，无法定义实数的实数的折射角 $\cos\theta'' \to \frac{k_z''}{k''} = \frac{\sqrt{k''^2 - k_x^2}}{k''}$

$$\cos\theta'' \to \frac{\sqrt{k''^2 - k_x^2}}{k''} = \frac{\sqrt{n_2^2k_0^2 - n_1^2k_0^2\sin^2\theta}}{n_2k_0} = \sqrt{1 - (\frac{\sin\theta}{n_{21}})^2} = \sqrt{1 - (\frac{\sin\theta}{\sin\theta_c})^2}$$

$$= i\alpha$$

其中$\alpha = \sqrt{(\frac{\sin\theta}{\sin\theta_c})^2 - 1}$

对于TE波(s波）

$$\frac{E'}{E} = \frac{Z_2\cos\theta - Z_1\cos\theta''}{Z_2\cos\theta + Z_1\cos\theta''} = \frac{Z_2\cos\theta - iZ_1\alpha}{Z_2\cos\theta + iZ_1\alpha} = \frac{e^{-i\phi}}{e^{i\phi}} = e^{-2i\phi}$$

其中$\phi = \arctan\frac{Z_1\alpha}{Z_2\cos\theta}$

全反射，反射波与入射波之间存在相位差

电磁波在界面的反射与折射

对于TM波(p波）

$$\frac{H'}{H}=\frac{Z_1\cos\theta-Z_2\cos\theta''}{Z_1\cos\theta+Z_2\cos\theta''}=\frac{Z_1\cos\theta-iZ_2\alpha}{Z_1\cos\theta+iZ_2\alpha}$$

全反射

对于折射波 $k_z''=k''\cos\theta''=in_2k_0\alpha=i\beta$

此时$\beta=n_2k_0\alpha=n_2k_0\sqrt{(\frac{\sin\theta}{\sin\theta_c})^2-1}$

$$\vec{E}_t''=\vec{E}''e^{ik_z''z}e^{i(k_x''x-\omega t)}=\vec{E}''e^{-\beta z}e^{i(k_x''x-\omega t)}$$

穿透深度 $$\delta=\frac{1}{\beta}=\frac{1}{n_2k_0\sqrt{(\frac{\sin\theta}{\sin\theta_c})^2-1}}=\frac{\lambda_1}{2\pi\sqrt{\sin^2\theta-{n_{21}}^2}}$$

波长量级

假设入射波为TE波（s波）

$$\vec{E}_t''=\hat{y}E''e^{ik_z''z}e^{i(k_xx-\omega t)}=\hat{y}E''e^{-\beta z}e^{i(k_xx-\omega t)}$$

电磁波在界面的反射与折射

$$H_t'' = \frac{1}{Z_2}(\hat{k}'' \times \vec{E}_t'') = \frac{E''}{Z_2 k''}(k_x\hat{z} - i\beta\hat{x})\, e^{-\beta z} e^{i(k_x x - \omega t)}$$

$$= \frac{E''}{\omega\mu_2}(k_x\hat{z} - i\beta\hat{x})\, e^{-\beta z} e^{i(k_x x - \omega t)}$$

$$\frac{1}{Z_2 k''} = \sqrt{\frac{\varepsilon_2}{\mu_2}}\frac{1}{\omega\sqrt{\mu_2\varepsilon_2}} = \frac{1}{\omega\mu_2}$$

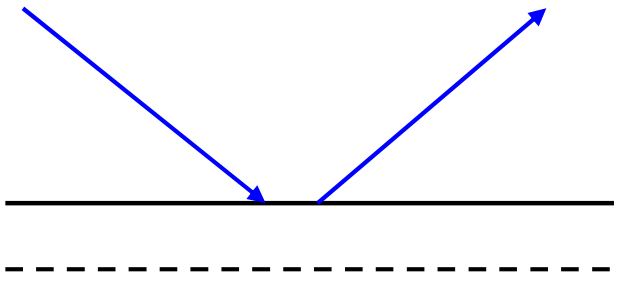
时间平均的能流$\langle\vec{S}\rangle = \langle\vec{E}\times\vec{H}\rangle = \frac{1}{2}\mathrm{Re}\{\vec{E}\times\vec{H}^*\}$

$$\vec{E}_t'' = \hat{y}E''e^{-\beta z}e^{i(k_x x - \omega t)}$$

$$\langle\vec{S}_x\rangle = \frac{E''^2 k_x}{2\omega\mu_2}e^{-2\beta z}$$

$$\langle\vec{S}_z\rangle = 0$$

能量沿着界面传输



Goos-Hänchen位移

电磁波在界面的反射与折射

例2：导体表面上的电磁波反射（垂直入射）。

解：对于良导体$\frac{\sigma}{\varepsilon_0\omega}\gg 1$，$\varepsilon_r=1+\frac{i\sigma}{\omega\varepsilon_0}\approx\frac{i\sigma}{\omega\varepsilon_0}$

$$r_s=\frac{E'}{E}=\frac{Z_2\cos\theta-Z_1\cos\theta''}{Z_2\cos\theta+Z_1\cos\theta''}=\frac{Z_2-Z_1}{Z_2+Z_1}$$

$$=\frac{1-\sqrt{\varepsilon_r}}{1+\sqrt{\varepsilon_r}}=-\frac{1-\sqrt{\frac{2\varepsilon_0\omega}{\sigma}}+i}{1+\sqrt{\frac{2\varepsilon_0\omega}{\sigma}}+i}$$

反射率

$$R_s=|r_s|^2=\frac{(1-\sqrt{\frac{2\varepsilon_0\omega}{\sigma}})^2+1}{(1+\sqrt{\frac{2\varepsilon_0\omega}{\sigma}})^2+1}\approx 1-2\sqrt{\frac{2\varepsilon_0\omega}{\sigma}}$$

$$Z_1=\sqrt{\frac{\mu_0}{\varepsilon_0}}$$

$$Z_2=\sqrt{\frac{\mu_0}{\varepsilon_0\varepsilon_r}}$$

$$\sqrt{\varepsilon_r}=(1+i)\sqrt{\frac{\sigma}{2\varepsilon_0\omega}}$$

- 当$\sigma\to\infty, R_s\to 1$,金属作为电磁波的反射镜

波导

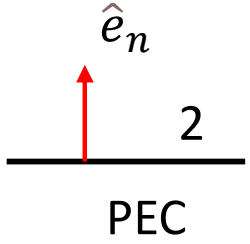
高频电磁能量的传输问题

- 恒定电流，求解电路方程（电压、电流、电阻，电容、电感等）例如双线传输、同轴传输线；
- 对于高频电磁波，能量存在于电磁场中，利用波导

1. 理想导体的边界条件

导体内的电磁波，由于趋肤效应，趋肤深度$\delta = \sqrt{\frac{2}{\omega\mu\sigma}}$，当导体为完美导体$\sigma \to \infty$，趋肤深度$\delta \to 0$, 导体内电场、磁场为0，电流电荷趋于在表面分布，导体构成电磁波的边界。

理想导体的电磁场边界条件



$$\begin{cases} \hat{e}_n \cdot (\vec{D}_2 - \vec{D}_1) = \sigma_f \\ \hat{e}_n \times (\vec{E}_2 - \vec{E}_1) = 0 \\ \hat{e}_n \cdot (\vec{B}_2 - \vec{B}_1) = 0 \\ \hat{e}_n \times (\vec{H}_2 - \vec{H}_1) = \vec{\alpha}_f \end{cases}$$

理想导体

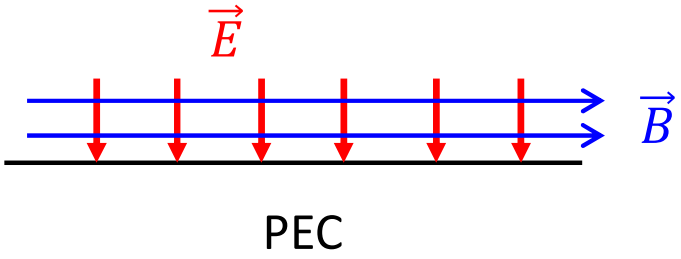
$$\begin{cases} \hat{e}_n \cdot \vec{D} = \sigma_f \\ \hat{e}_n \times \vec{E} = 0 \\ \hat{e}_n \cdot \vec{B} = 0 \\ \hat{e}_n \times \vec{H} = \vec{\alpha}_f \end{cases}$$

$\hat{e}_n \times \vec{E} = 0$

- 切向电场为0，电场垂直于界面

$\hat{e}_n \cdot \vec{B} = 0$

- 法线磁场为0，磁场平行于界面



2. 波导场方程

无源真空中的电磁场

$$\begin{cases} (\nabla^2 + k^2)\vec{E}(\vec{x}) = 0 \\ \nabla \cdot \vec{E} = 0 \\ \vec{B} = \dfrac{1}{i\omega}\nabla \times \vec{E} \end{cases} \quad \text{或} \quad \begin{cases} (\nabla^2 + k^2)\vec{B}(\vec{x}) = 0 \\ \nabla \cdot \vec{B} = 0 \\ \vec{E} = \dfrac{i}{\omega\mu\varepsilon}\nabla \times \vec{B} \end{cases}$$

设电磁场沿着z方向传输，考虑试解：

$$\begin{pmatrix} \vec{E} \\ \vec{B} \end{pmatrix} = \begin{pmatrix} \vec{E}_0(x,y) \\ \vec{B}_0(x,y) \end{pmatrix} e^{ik_z z} e^{-i\omega t}$$

代入亥姆霍兹方程，可得：

$$k_c^2 = \frac{\omega^2}{c^2} - k_z^2$$

$$\left[\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} - k_z^2\right) + \frac{\omega^2}{c^2}\right]\begin{pmatrix} \vec{E}_0(x,y) \\ \vec{B}_0(x,y) \end{pmatrix} = 0 \qquad \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + k_c^2\right)\begin{pmatrix} \vec{E}_0(x,y) \\ \vec{B}_0(x,y) \end{pmatrix} = 0$$

波导

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+k_c^2\right)\begin{pmatrix}\vec{E}_0(x,y)\\ \vec{B}_0(x,y)\end{pmatrix}=0 \qquad \text{其中 } k_c^2=\frac{\omega^2}{c^2}-k_z^2$$

受限空间中的电磁波不一定是横波，可能由六个分量$E_{0x},E_{0y},E_{0z},B_{0x},B_{0y}$和$B_{0z}$,然而这些分量并不是独立的，可以先推导这些分量之间的联系

$$\nabla\times\vec{E}=-\frac{\partial\vec{B}}{\partial t}$$

$$\frac{\partial}{\partial z}\to ik_z$$

$$\begin{cases}\dfrac{\partial E_{0y}}{\partial z}-\dfrac{\partial E_{0z}}{\partial y}=-i\omega B_{0x}\\ \dfrac{\partial E_{0x}}{\partial z}-\dfrac{\partial E_{0z}}{\partial x}=i\omega B_{0y}\\ \dfrac{\partial E_{0y}}{\partial x}-\dfrac{\partial E_{0x}}{\partial y}=i\omega B_{0z}\end{cases} \qquad \begin{cases}ik_zE_{0y}-\dfrac{\partial E_{0z}}{\partial y}=-i\omega B_{0x}\\ ik_zE_{0x}-\dfrac{\partial E_{0z}}{\partial x}=i\omega B_{0y}\\ \dfrac{\partial E_{0y}}{\partial x}-\dfrac{\partial E_{0x}}{\partial y}=i\omega B_{0z}\end{cases}$$

波导

$$\nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t} \qquad \nabla \times \vec{B} = \frac{1}{c^2}\frac{\partial \vec{E}}{\partial t} = \frac{-i\omega}{c^2}\vec{E}$$

$$\begin{cases} \dfrac{\partial B_{0y}}{\partial z} - \dfrac{\partial B_{0z}}{\partial y} = \dfrac{i\omega}{c^2}E_{0x} \\ \dfrac{\partial B_{0x}}{\partial z} - \dfrac{\partial B_{0z}}{\partial x} = \dfrac{-i\omega}{c^2}E_{0y} \\ \dfrac{\partial B_{0y}}{\partial x} - \dfrac{\partial B_{0x}}{\partial y} = \dfrac{-i\omega}{c^2}E_{0z} \end{cases} \qquad \frac{\partial}{\partial z} \to ik_z \qquad \begin{cases} ik_z B_{0y} - \dfrac{\partial B_{0z}}{\partial y} = \dfrac{i\omega}{c^2}E_{0x} \\ ik_z B_{0x} - \dfrac{\partial B_{0z}}{\partial x} = \dfrac{-i\omega}{c^2}E_{0y} \\ \dfrac{\partial B_{0y}}{\partial x} - \dfrac{\partial B_{0x}}{\partial y} = \dfrac{-i\omega}{c^2}E_{0z} \end{cases}$$

$$\begin{cases} ik_z E_{0y} - \dfrac{\partial E_{0z}}{\partial y} = -i\omega B_{0x} \\ ik_z E_{0x} - \dfrac{\partial E_{0z}}{\partial x} = i\omega B_{0y} \\ \dfrac{\partial E_{0y}}{\partial x} - \dfrac{\partial E_{0x}}{\partial y} = i\omega B_{0z} \end{cases}$$

- E_{0x}，B_{0y}，B_{0x}，E_{0y}可以通过E_{0z}和B_{0z}表示出来，若E_{0z}, B_{0z}确定，其值可以求解

波导

$$\begin{cases} E_{0x} = \dfrac{i}{k_c^2}\left[k_z \dfrac{\partial E_{0z}}{\partial x} + k_0 \dfrac{\partial (cB_{0z})}{\partial y}\right] \\ E_{0y} = \dfrac{i}{k_c^2}\left[k_z \dfrac{\partial E_{0z}}{\partial y} - k_0 \dfrac{\partial (cB_{0z})}{\partial x}\right] \\ B_{0x} = \dfrac{i}{ck_c^2}\left[-k_0 \dfrac{\partial E_{0z}}{\partial y} + k_z \dfrac{\partial (cB_{0z})}{\partial x}\right] \\ B_{0y} = \dfrac{i}{ck_c^2}\left[k_0 \dfrac{\partial E_{0z}}{\partial x} + k_z \dfrac{\partial (cB_{0z})}{\partial y}\right] \end{cases}$$

其中$k_0 = \dfrac{\omega}{c}$

因此波导场方程可以简化为两个，即：

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + k_c^2\right)\begin{pmatrix} E_{0z} \\ B_{0z} \end{pmatrix} = 0$$

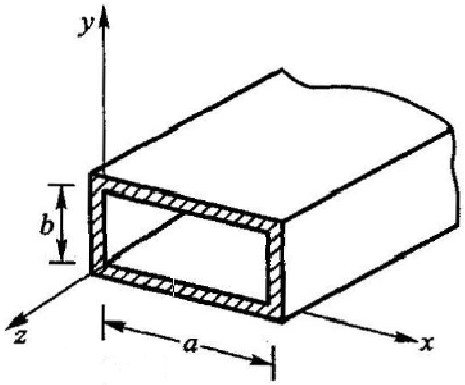
- 若$E_{0z} = 0$，$B_{0z} = 0$，称为TEM模，横电磁模

波导模式为该方程的解，分为两类：

（1） $B_{0z} \neq 0$， $E_{0z} = 0$，称为TE模，横电模

（2） $E_{0z} \neq 0$， $B_{0z} = 0$，称为TM模，横磁模

3. 矩形波导中的电磁波



先考虑横电模（TE模）（$B_{0z} \neq 0$， $E_{0z} = 0$）

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + k_c^2\right) B_{0z} = 0$$

满足边界条件 $\hat{e}_n \times \vec{E} = 0$， $\hat{e}_n \cdot \vec{B} = 0$，即：

$$\begin{cases} E_{0x} = \dfrac{i}{k_c^2}\left[k_z \dfrac{\partial E_{0z}}{\partial x} + k_0 \dfrac{\partial (cB_{0z})}{\partial y}\right] \\ E_{0y} = \dfrac{i}{k_c^2}\left[k_z \dfrac{\partial E_{0z}}{\partial y} - k_0 \dfrac{\partial (cB_{0z})}{\partial x}\right] \\ B_{0x} = \dfrac{i}{ck_c^2}\left[-k_0 \dfrac{\partial E_{0z}}{\partial y} + k_z \dfrac{\partial (cB_{0z})}{\partial x}\right] \\ B_{0y} = \dfrac{i}{ck_c^2}\left[k_0 \dfrac{\partial E_{0z}}{\partial x} + k_z \dfrac{\partial (cB_{0z})}{\partial y}\right] \end{cases}$$

$$E_{0x}|_{y=0,b} = E_{0y}|_{x=0,a} = 0$$

$$B_{0x}|_{x=0,a} = B_{0y}|_{y=0,b} = 0$$

$$\left.\frac{\partial B_{0z}}{\partial y}\right|_{y=0,b} = \left.\frac{\partial B_{0z}}{\partial x}\right|_{x=0,a} = 0$$

波导

波方程的通解：

$$B_{0z} = B_0(\cos k_x x,\ \sin k_x x)(\cos k_y y,\ \sin k_y y)$$

$$-k_x^2 - k_y^2 + k_c^2 = 0$$

其中 $k_c^2 = \frac{\omega^2}{c^2} - k_z^2$

由边界条件：$\left.\frac{\partial B_{0z}}{\partial x}\right|_{x=0,a} = 0$，只能选择$\cos k_x x$，并且$k_x a = m\pi$，由此可得：

$$k_x = \frac{m\pi}{a}$$

由边界条件：$\left.\frac{\partial B_{0z}}{\partial y}\right|_{y=0,b} = 0$，只能选择$\cos k_y y$，并且$k_y b = 2m\pi$，由此可得：

$$k_y = \frac{n\pi}{b}$$

综合得到：

$$B_{0z} = B_0 \cos k_x x \cos k_y x = B_0 \cos\frac{m\pi}{a} x \cos\frac{n\pi}{b} y$$

其中有：

$$k_c^2 = k_x^2 + k_y^2 = \pi^2\left[\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2\right]$$

波导

由场的分量关系，可以其余的场分量

$$\begin{cases} E_{0x}=\dfrac{ik_0c}{k_c^2}\dfrac{\partial B_{0z}}{\partial y}=-i\left(\dfrac{n\pi}{b}\right)\dfrac{k_0c}{k_c^2}B_0\cos\dfrac{m\pi}{a}x\sin\dfrac{n\pi}{b}y \\ E_{0y}=-\dfrac{ik_0c}{k_c^2}\dfrac{\partial B_{0z}}{\partial x}=i\left(\dfrac{m\pi}{a}\right)\dfrac{k_0c}{k_c^2}B_0\sin\dfrac{m\pi}{a}x\cos\dfrac{n\pi}{b}y \\ E_{0z}=0 \\ B_{0x}=\dfrac{ik_z}{k_c^2}\dfrac{\partial B_{0z}}{\partial x}=-\dfrac{ik_z}{k_c^2}\left(\dfrac{m\pi}{a}\right)B_0\sin\dfrac{m\pi}{a}x\cos\dfrac{n\pi}{b}y \\ B_{0y}=\dfrac{ik_z}{k_c^2}\dfrac{\partial B_{0z}}{\partial y}=-\dfrac{ik_z}{k_c^2}\left(\dfrac{n\pi}{b}\right)B_0\cos\dfrac{m\pi}{a}x\sin\dfrac{n\pi}{b}y \\ B_{0z}=B_0\cos\dfrac{m\pi}{a}x\cos\dfrac{n\pi}{b}y \end{cases}$$

$$\begin{cases} E_{0x}=\dfrac{i}{k_c^2}\left[k_z\dfrac{\partial E_{0z}}{\partial x}+k_0\dfrac{\partial(cB_{0z})}{\partial y}\right] \\ E_{0y}=\dfrac{i}{k_c^2}\left[k_z\dfrac{\partial E_{0z}}{\partial y}-k_0\dfrac{\partial(cB_{0z})}{\partial x}\right] \\ B_{0x}=\dfrac{i}{ck_c^2}\left[-k_0\dfrac{\partial E_{0z}}{\partial y}+k_z\dfrac{\partial(cB_{0z})}{\partial x}\right] \\ B_{0y}=\dfrac{i}{ck_c^2}\left[k_0\dfrac{\partial E_{0z}}{\partial x}+k_z\dfrac{\partial(cB_{0z})}{\partial y}\right] \end{cases}$$

- 已经求到传播电磁波色散关系、场分量，问题得到解决。

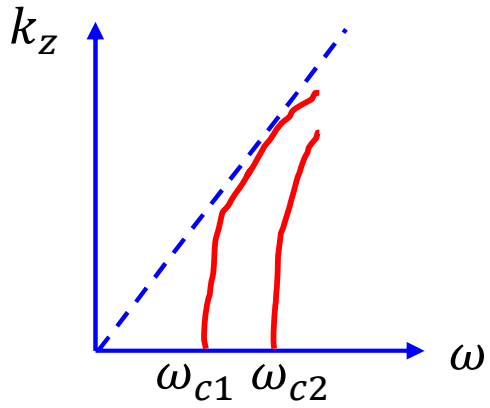
波导

讨论：

(1) 波模式的色散关系

$$k_z=\sqrt{k_0^2-k_c^2}=\frac{\omega}{c}\sqrt{1-(\frac{ck_c}{\omega})^2}=\frac{\omega}{c}\sqrt{1-\frac{\omega_c^2}{\omega^2}}$$

其中$\omega_c=ck_c=\pi c\sqrt{\left(\frac{m}{a}\right)^2+\left(\frac{n}{b}\right)^2}$



(2) 截止频率

当$\omega<\omega_c$，k_z为虚数，模式无法传播；当$\omega>\omega_c$，k_z为实数，模式可以传播，

因此截止频率为$\omega_c=\pi c\sqrt{\left(\frac{m}{a}\right)^2+\left(\frac{n}{b}\right)^2}$

(3) 波模

(m,n)取不同的值，波导截面的场具有不同的分布，每一种(m,n)称之为一模式，记为TE_{mn}, TE的最低阶模式：当$a<b$, TE_{01}; 当$a>b$, TE_{10}

若$a > b$， 截止频率$\omega_{c,10} = \frac{\pi c}{a}$, 截止线频率$f = \frac{c}{2a}$, 截止波长$\lambda = \frac{2\pi c}{\omega_{c,10}} = 2a$

(4) 能流 $\langle \vec{S} \rangle = \frac{1}{2}\mathrm{Re}\{\vec{E} \times \vec{H}^*\}$

- (E_x, H_y)同相位，能流沿着z方向传输；
- $(E_x, H_z), (E_y, H_z)$相位差为$\frac{\pi}{2}$， 能流在x,y方向无法传输

(5) 波速

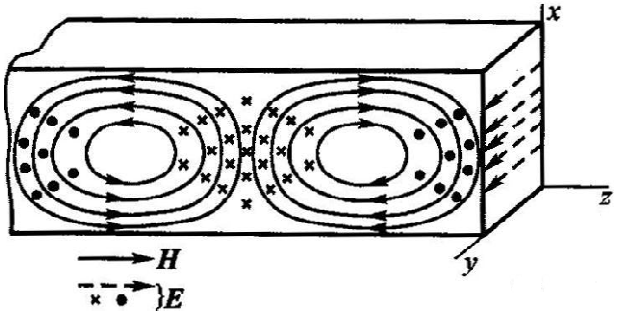
相速度$v = \frac{\omega}{k_z} = \frac{\omega}{\frac{\omega}{c}\sqrt{1-\frac{\omega_c^2}{\omega^2}}} = \frac{c}{\sqrt{1-\frac{\omega_c^2}{\omega^2}}} > c$

群速度$v_g = \frac{d\omega}{dk_z} = \frac{1}{dk_z/d\omega} = c\sqrt{1-\frac{\omega_c^2}{\omega^2}} < c$

(6) 考虑TE_{10}模式 $k_x = \frac{\pi}{a}$, $k_y = 0$, $k_c = \frac{\pi}{a}$

波导

$$\begin{cases} E_{0x}=0 \\ E_{0y}=i\dfrac{\omega a}{\pi}B_0\sin\dfrac{\pi}{a}x \\ E_{0z}=0 \\ B_{0x}=-\dfrac{iak_z}{\pi}B_0\sin\dfrac{\pi}{a}x \\ B_{0y}=0 \\ B_{0z}=B_0\cos\dfrac{\pi}{a}x \end{cases}$$



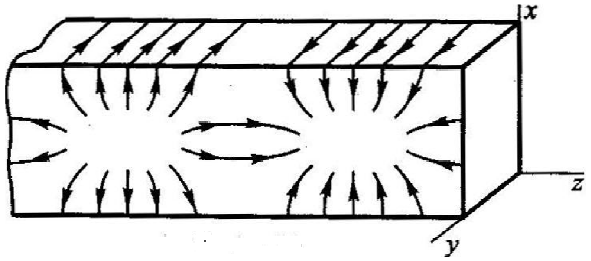
求出磁场后，由边界条件可得出管壁上的电流分布　　$\hat{e}_n\times\vec{H}=\vec{\alpha}_f$

在$x=a$面上，电流密度$\vec{\alpha}_f\big|_{x=a}=-\hat{x}\times\mu_0B_0\cos\dfrac{\pi}{a}x\,e^{ik_zz}\hat{z}\Big|_{x=a}=H_0e^{ik_zz}\hat{y}$

在y= b面上，电流密度$\vec{\alpha}_f\big|_{y=b}=-\hat{y}\times(\dfrac{iak_z}{\pi}B_0\sin\dfrac{\pi}{a}x\,e^{ik_zz}\hat{x}+\mu_0B_0\cos\dfrac{\pi}{a}x\,e^{ik_zz}\hat{z})\Big|_{y=b}$

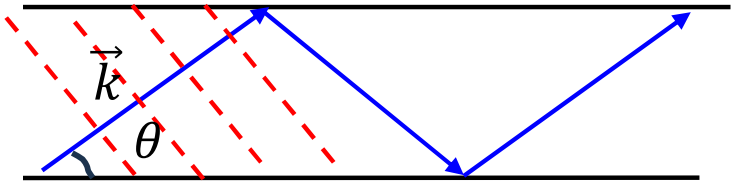
$=\dfrac{iak_z}{\pi}H_0\sin\dfrac{\pi}{a}x\,e^{ik_zz}\hat{z}-H_0\cos\dfrac{\pi}{a}x\,e^{ik_zz}\hat{x}$

- 波导窄边上没有纵向电流，电流是横过窄边
- 在波导宽边中线上，横向电流为零



(7) 波导管内电磁波传播的另一种解释

在x，y方向上形成驻波，$m\frac{\lambda_x}{2}=a \quad n\frac{\lambda_y}{2}=b$



$$k_y=\frac{2\pi}{\lambda_y}=\frac{n\pi}{b} \qquad k_x=\frac{2\pi}{\lambda_x}=\frac{m\pi}{a}$$

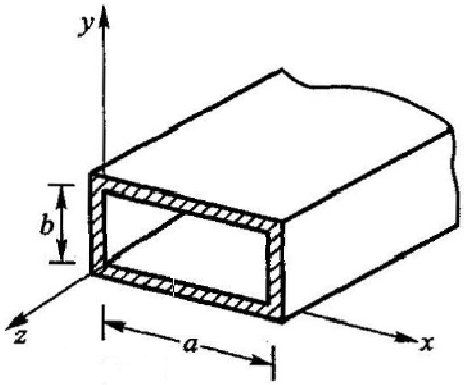
入射波波矢 $\vec{k}=k_x\hat{x}+k_y\hat{y}+k_z\hat{z}=\frac{m\pi}{a}\hat{x}+\frac{n\pi}{b}\hat{y}+k_z\hat{z}$

$$\omega_{m,n}=\pi c\sqrt{\left(\frac{m}{a}\right)^2+\left(\frac{n}{b}\right)^2}$$

$$|\vec{k}|=\frac{\omega}{c}=\sqrt{(\frac{m\pi}{a})^2+(\frac{n\pi}{b})^2+k_z^2}=\sqrt{\frac{1}{c^2}\left[\pi^2c^2\left(\frac{m}{a}\right)^2+\left(\frac{n}{b}\right)^2\right]+k_z^2}$$

$$=\sqrt{\frac{\omega_{m,n}^2}{c^2}+k_z^2} \qquad k_z=\frac{1}{c}\sqrt{\omega^2-\omega_{m,n}^2}$$

入射角θ $\cos\theta=\frac{k_z}{|\vec{k}|}=\sqrt{1-\frac{\omega_{m,n}^2}{\omega^2}}$ 若θ有解，必须$\omega>\omega_{m,n}$



考虑横磁模（TM模）（$E_{0z} \neq 0$，$B_{0z} = 0$）

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + k_c^2\right) E_{0z} = 0$$

满足边界条件 $\hat{e}_n \times \vec{E} = 0$，$\hat{e}_n \cdot \vec{B} = 0$，即：

$$E_{0z}|_{x=0,a} = E_{0z}|_{y=0,b} = 0$$

波方程的通解：

$$E_{0z} = E_0(\cos k_x x,\ \sin k_x x)(\cos k_y y,\ \sin k_y y)$$

考虑边界条件取$\sin k_x x$,$\sin k_y y$ 项,并且有$\sin k_x a = 0$,$\sin k_y b = 0$

$$k_x = \frac{m\pi}{a} \qquad k_y = \frac{n\pi}{b} \qquad k_c^2 = k_x^2 + k_y^2 = \pi^2\left[\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2\right]$$

$$\Rightarrow \quad E_{0z} = E_0 \sin k_x x \sin k_y y = E_0 \sin\frac{m\pi}{a} x \sin\frac{n\pi}{b} y$$

代入场分量关系公式，可得：

$$\begin{cases} E_{0x}=\dfrac{ik_z}{k_c^2}\dfrac{\partial E_{0z}}{\partial x}=i\dfrac{m\pi}{a}\dfrac{k_z}{k_c^2}E_0\cos\dfrac{m\pi}{a}x\sin\dfrac{n\pi}{b}y \\ E_{0y}=\dfrac{ik_z}{k_c^2}\dfrac{\partial E_{0z}}{\partial y}=i\dfrac{n\pi}{b}\dfrac{k_z}{k_c^2}E_0\sin\dfrac{m\pi}{a}x\cos\dfrac{n\pi}{b}y \\ E_{0z}=E_0\sin\dfrac{m\pi}{a}x\sin\dfrac{n\pi}{b}y \\ B_{0x}=-\dfrac{ik_0}{ck_c^2}\dfrac{\partial E_{0z}}{\partial y}=-i\dfrac{k_0}{ck_c^2}\dfrac{n\pi}{b}E_0\sin\dfrac{m\pi}{a}x\cos\dfrac{n\pi}{b}y \\ B_{0y}=\dfrac{ik_0}{ck_c^2}\dfrac{\partial E_{0z}}{\partial x}=\dfrac{ik_0}{ck_c^2}\dfrac{m\pi}{a}E_0\cos\dfrac{m\pi}{a}x\sin\dfrac{n\pi}{b}y \\ B_{0z}=0 \end{cases}$$

$$\begin{cases} E_{0x}=\dfrac{i}{k_c^2}[k_z\dfrac{\partial E_{0z}}{\partial x}+k_0\dfrac{\partial(cB_{0z})}{\partial y}] \\ E_{0y}=\dfrac{i}{k_c^2}[k_z\dfrac{\partial E_{0z}}{\partial y}-k_0\dfrac{\partial(cB_{0z})}{\partial x}] \\ B_{0x}=\dfrac{i}{ck_c^2}[-k_0\dfrac{\partial E_{0z}}{\partial y}+k_z\dfrac{\partial(cB_{0z})}{\partial x}] \\ B_{0y}=\dfrac{i}{ck_c^2}[k_0\dfrac{\partial E_{0z}}{\partial x}+k_z\dfrac{\partial(cB_{0z})}{\partial y}] \end{cases}$$

波导

讨论：

(1) 能流 $\langle \vec{S} \rangle = \frac{1}{2}\mathrm{Re}\{\vec{E} \times \vec{H}^*\}$

- (E_x, H_y)同相位，能流沿着z方向传输；
- $(E_x, H_z), (E_y, H_z)$相位差为$\frac{\pi}{2}$，能流在x,y方向无法传输

(2) 基模，m, n都不能为0，否则$E_{0z} = 0$，所有的场量都为0，因此最低级模式为TM_{11}

(3) 截止频率

$$\omega_c = k_c c = \pi c \sqrt{\frac{1}{a^2} + \frac{1}{b^2}}$$

$$\frac{\omega_{c,TM}}{\omega_{c,TE}} = \frac{\pi c \sqrt{\frac{1}{a^2} + \frac{1}{b^2}}}{\frac{\pi c}{a}} = \sqrt{1 + \frac{a^2}{b^2}} > 1$$

谐振腔

电磁波由特定谐振频率的电路元件激发，在低频情况，采用LC回路，谐振频率$\frac{1}{2\pi\sqrt{LC}}$；在高频中，金属壁面谐振腔；光学中的谐振腔是由反射镜组成的光学谐振腔

由完美电导体（PEC）组成的封闭腔体，腔内的电磁场满足

$$\begin{cases} \nabla\cdot\vec{E}=0 \\ \nabla\times\vec{E}=i\omega\vec{B} \\ \nabla\cdot\vec{B}=0 \\ \nabla\times\vec{B}=-\frac{i\omega}{c^2}\vec{E} \end{cases}$$

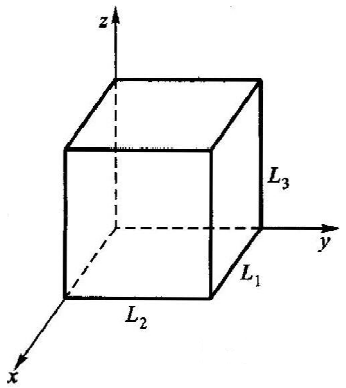
边界上满足 $\begin{cases} \hat{e}_n\times\vec{E}=0 \\ \hat{e}_n\cdot\vec{B}=0 \end{cases}$

满足场方程+边界条件的解。

谐振腔

考虑矩形谐振腔$L_1 \times L_2 \times L_3$，等效为截面为$L_1 \times L_2$的波导，

在$z=0, L_3$端面上加上PEC边界条件



无端面的波导模式，取$TE_{m,n}$,则

$$B_z = B_0 \cos\frac{m\pi}{L_1} x \cos\frac{n\pi}{L_2} y\, e^{ik_z z} e^{-i\omega t}$$

由于端面的存在，电磁波会发生反射，此时的电磁波为：

$$B_z = (B_0 \cos\frac{m\pi}{L_1} x \cos\frac{n\pi}{L_2} y\, e^{ik_z z} + B_0' \cos\frac{m\pi}{L_1} x \cos\frac{n\pi}{L_2} y\, e^{-ik_z z}) e^{-i\omega t}$$

必须满足边界条件：$B_{0z}|_{z=0,L_3} = 0$

由 $B_{0z}|_{z=0} = 0$ $B_0 + B_0' = 0$

由 $B_{0z}|_{z=L_3} = 0$ $B_0 e^{ik_z L_3} + B_0' e^{-ik_z L_3} = 0$

$$\sin k_z L_3 = 0 \qquad k_z = \frac{p\pi}{L_3}$$

谐振腔

腔内电磁波的解为：

$$B_z = B_0(\cos\frac{m\pi}{L_1}x\cos\frac{n\pi}{L_2}y\,e^{i\frac{p\pi}{L_3}z} - \cos\frac{m\pi}{L_1}x\cos\frac{n\pi}{L_2}y\,e^{-i\frac{p\pi}{L_3}z})e^{-i\omega t}$$

$$= 2iB_0\cos\frac{m\pi}{L_1}x\cos\frac{n\pi}{L_2}y\sin\frac{p\pi}{L_3}z\,e^{-i\omega t} = B_{0z}e^{-i\omega t}$$

$$B_{0z} = 2iB_0\cos\frac{m\pi}{L_1}x\cos\frac{n\pi}{L_2}y\sin\frac{p\pi}{L_3}z$$

$$\begin{cases} E_{0x} = \dfrac{i}{k_c^2}[k_z\dfrac{\partial E_{0z}}{\partial x} + k_0\dfrac{\partial(cB_{0z}}{\partial y} \\ E_{0y} = \dfrac{i}{k_c^2}[k_z\dfrac{\partial E_{0z}}{\partial y} - k_0\dfrac{\partial(cB_{0z}}{\partial x} \\ B_{0x} = \dfrac{i}{ck_c^2}[-k_0\dfrac{\partial E_{0z}}{\partial y} + k_z\dfrac{\partial(cB_{0}}{\partial x} \\ B_{0y} = \dfrac{i}{ck_c^2}[k_0\dfrac{\partial E_{0z}}{\partial x} + k_z\dfrac{\partial(cB_{0}}{\partial y} \end{cases}$$

$$\begin{cases} E_{0x} = \dfrac{i}{k_c^2}k_0\dfrac{\partial(cB_{0z})}{\partial y} = \dfrac{2\omega B_0}{k_c^2}\dfrac{n\pi}{L_2}\cos\dfrac{m\pi}{L_1}x\sin\dfrac{n\pi}{L_2}y\sin\dfrac{p\pi}{L_3}z \\ E_{0y} = -\dfrac{i}{k_c^2}k_0\dfrac{\partial(cB_{0z})}{\partial x} = -\dfrac{2\omega B_0}{k_c^2}\dfrac{m\pi}{L_1}\sin\dfrac{m\pi}{L_1}x\cos\dfrac{n\pi}{L_2}y\sin\dfrac{p\pi}{L_3}z \\ E_{0z} = 0 \\ B_{0x} = \dfrac{i}{ck_c^2}k_z\dfrac{\partial(cB_{0z})}{\partial x} = \dfrac{1}{k_c^2}\dfrac{\partial}{\partial z}\left(\dfrac{\partial B_{0z}}{\partial x}\right) = -\dfrac{2iB_0}{k_c^2}\dfrac{m\pi}{L_1}\dfrac{p\pi}{L_3}\sin\dfrac{m\pi}{L_1}x\cos\dfrac{n\pi}{L_2}y\cos\dfrac{p\pi}{L_3}z \\ B_{0y} = \dfrac{i}{ck_c^2}k_z\dfrac{\partial(cB_{0z})}{\partial y} = \dfrac{1}{k_c^2}\dfrac{\partial}{\partial z}\left(\dfrac{\partial B_{0z}}{\partial y}\right) = -\dfrac{2iB_0}{k_c^2}\dfrac{n\pi}{L_2}\dfrac{p\pi}{L_3}\cos\dfrac{m\pi}{L_1}x\sin\dfrac{n\pi}{L_2}y\cos\dfrac{p\pi}{L_3}z \\ B_{0z} = 2iB_0\cos\dfrac{m\pi}{L_1}x\cos\dfrac{n\pi}{L_2}y\sin\dfrac{p\pi}{L_3}z \end{cases}$$

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讨论：

（1）(E_x, E_y)与(B_x, B_y)之间差$\pi/2$相位，各个方向平均能流为0，表面能流的局域；

（2） E_{0x}， E_{0y}满足边界条件$E_{0x}|_{z=0,L_3} = 0$, $E_{0y}|_{z=0,L_3} = 0$,在该端面上电场无切向电场；

（3）色散关系。由$k_z = \frac{p\pi}{L_3}$， $k_z^2 = k_0^2 - k_c^2 = k_0^2 - \pi^2[\left(\frac{m}{L_1}\right)^2 + \left(\frac{n}{L_2}\right)^2]$, 可得：

$\pi^2\left[\left(\frac{m}{L_1}\right)^2 + \left(\frac{n}{L_2}\right)^2 + \left(\frac{p}{L_3}\right)^2\right] = k_0^2 = \frac{\omega^2}{c^2}$, 谐振频率$\omega = \pi c\sqrt{\left(\frac{m}{L_1}\right)^2 + \left(\frac{n}{L_2}\right)^2 + \left(\frac{p}{L_3}\right)^2}$, 谐振波长$\lambda = \frac{2\pi c}{\omega} = \frac{2}{\sqrt{\left(\frac{m}{L_1}\right)^2 + \left(\frac{n}{L_2}\right)^2 + \left(\frac{p}{L_3}\right)^2}}$

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(4) (m,n,p)中至多只有1个为0，故最低阶模式为$(110),(101)$或者(011)；

(5) 若$L_1 > L_2 > L_3$，最低阶波模谐振频率为$\omega_{110} = \pi c\sqrt{\left(\frac{1}{L_1}\right)^2 + \left(\frac{1}{L_2}\right)^2}$，对应的线频率为$f = \frac{\omega_{110}}{2\pi} = \frac{c}{2}\sqrt{\left(\frac{1}{L_1}\right)^2 + \left(\frac{1}{L_2}\right)^2}$，波长为$\lambda_{110} = \frac{2}{\sqrt{\left(\frac{1}{L_1}\right)^2 + \left(\frac{1}{L_2}\right)^2}}$。若$L_1 = L_2 = a$，$\lambda_{110} = \sqrt{2}a$（最大可以容纳波长）

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无端面的波导模式，取$TM_{m,n}$,则

$$E_z = E_0 \sin\frac{m\pi}{L_1}x \sin\frac{n\pi}{L_2}y\, e^{ik_z z}e^{-i\omega t}$$

由于端面的存在，电磁波会发生反射，此时的电磁波为：

$$E_z = (E_0 \sin\frac{m\pi}{L_1}x \sin\frac{n\pi}{L_2}y\, e^{ik_z z} + E_0' \sin\frac{m\pi}{L_1}x \sin\frac{n\pi}{L_2}y\, e^{-ik_z z})e^{-i\omega t}$$

必须满足边界条件： $\nabla \cdot \vec{E} = 0$, 即$\left.\frac{\partial E_z}{\partial z}\right|_{z=0,L_3} = 0$

由 $\left.\frac{\partial E_z}{\partial z}\right|_{z=0} = 0$ $E_0 - E_0' = 0$

由 $\left.\frac{\partial E_z}{\partial z}\right|_{z=L_3} = 0$ $\sin k_z L_3 = 0$ $k_z = \frac{p\pi}{L_3}$

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腔内电磁波的解为：

$$E_z = (E_0 \sin\frac{m\pi}{L_1}x \sin\frac{n\pi}{L_2}y\, e^{ik_z z} + E_0' \sin\frac{m\pi}{L_1}x \sin\frac{n\pi}{L_2}y\, e^{-ik_z z})e^{-i\omega t}$$

$$= E_0 \sin\frac{m\pi}{L_1}x \sin\frac{n\pi}{L_2}y\,(e^{i\frac{p\pi}{L_3}z} + e^{-i\frac{p\pi}{L_3}z})e^{-i\omega t}$$

$$= 2E_0 \sin\frac{m\pi}{L_1}x \sin\frac{n\pi}{L_2}y \cos\frac{p\pi}{L_3}z\, e^{-i\omega t}$$